

A unified approach to Bermudan and Barrier options under stochastic volatility models with jumps

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Abstract

Many financial assets, such as currencies, commodities, and equity stocks, exhibit both jumps and stochastic volatility, which are especially prominent in the market after the financial crisis. Some strategic decision making problems also involve American-style options. In this paper, we develop a novel, fast and accurate method for pricing American and barrier options in regime switching jump diffusion models. By blending regime switching models and Markov chain approximation techniques in the Fourier domain, we provide a *unified* approach to price Bermudan, American options and barrier options under general stochastic volatility models with jumps. The models considered include Heston, Hull-White, Stein-Stein, Scott, the 3/2 model, and the recently proposed 4/2 model and the α -Hypergeometric model with general jump amplitude distributions in the return process. Applications include the valuation of discretely monitored contracts as well as continuously monitored contracts common in the foreign exchange markets. Numerical results are provided to demonstrate the accuracy and efficiency of the proposed method.

Keywords: American options, barrier options, stochastic volatility, regime switching, jump diffusion, frame projection

AMS subject classifications: 91G80, 93E11, 93E20

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1 Introduction

American call (put) options give the holder the right to buy (sell) an underlying asset at a predetermined strike price at any point in time until the expiration date. According to Barraclough and Whaley (2012) and Jensen and Pedersen (2016), a large portion of options traded on the market are American-style options. Some strategic decision making problems also involve American-style real options, see Copeland and Tufano (2004) and Battauz et al. (2014). However, even under the simple Black and Scholes model based on geometric Brownian motions (GBM), closed-form solutions do not exist for American options outside of the infinite series expansions as in Zhu (2006). As a result, numerical methods must be employed to price and hedge these financial derivatives. The computational approaches can be categorized as Monte Carlo methods developed in Broadie and Glasserman (1997); Longstaff and Schwartz (2001); Lindset and Lund (2007); Fujiwara and Kijima (2007), lattice method of Cox et al. (1979), numerical partial differential equation (PDE) techniques in Ikonen and Toivanen (2004); Muthuraman (2008), randomization techniques developed in Carr (1998), integral equation methods of Myneni (1992); Carr et al. (1992); Jacka (1991), finite element method in Zhu and Chen (2013), radial basis method in Ballestra and Pacelli (2013), expansion method in Medvedev and Scaillet (2010), and the moving boundary approach in Muthuraman (2008). For an overview of the literature on pricing American options, we refer the reader to Broadie and Detemple (2004), Detemple (2005), Detemple and Emmerling (2009) and references therein.

The previous literature mentioned above considers American options under the constant volatility model, i.e. the Black and Scholes model. However, empirical evidence reveals the stochastic nature of the volatility. Earlier studies in Scott (1987) further confirm the random change of volatility over time, and Rubinstein (1994) provides empirical evidence suggesting that the constant volatility does not hold for options written on the S&P 500 index. Bates (1996) provided further evidence of the existence of jumps in the underlying index. As a result, several extensions have been proposed to account for more realistic descriptions for volatility dynamics. Among the most popular extensions are stochastic volatility models of Hull and White (1990), Heston (1993), Stein and Stein (1991), Scott (1987), the 3/2 model (see Carr and Sun (2007)), and recently the 4/2 model of Grasselli (2016) and the α -Hypergeometric model of Da Fonseca and Martini (2016).

While the pricing of European options under stochastic volatility models has been extensively studied in the literature, relatively little attention has been paid to the problem of American options under stochastic volatility models. Approaches in the literature include the partial differential equation (PDE) with penalty method of Zvan et al. (1998), the exercise-policy improvement procedure of Chockalingam and Muthuraman (2011), the sequential Monte Carlo

method developed in Rambharat and Brockwell (2010), and a nonparametric approach utilized in Broadie et al. (2000). For Heston model, Ikonen and Toivanen (2008) consider the problem of pricing American put options using five numerical methods, and Zhylyevskyy (2010) develops a fast Fourier transform approach to price American options by extending the Geske-Johnson compound option scheme.

So far the literature discussed above is limited to stochastic volatility models *without jumps*. On the other hand, empirical studies reveal that there exist sudden rapid movements in the mid quotes of stock prices, i.e. jumps during the trading period, as documented in empirical studies by Bates (1996), Carr and Wu (2003), Lee and Hannig (2010), etc. The recent financial crisis highlights that event risks such as a market crash may be significant, e.g. the *flash crash* on May 6th, 2010. Motivated from the need to model random volatility fluctuations and the existence of jumps, stochastic volatility models with jumps have been developed. They improve the pricing and hedging performance of traditional jump diffusion models (see Merton (1976)), and in particular, account for certain imperfections, such as the volatility smile. In contrast to the case of diffusion models, very few studies are known for American options in stochastic volatility models with jumps.

In a recent paper, Salmi et al. (2014) develop a (two-step implicit-explicit) IMEX scheme to price American put options in Heston model with simultaneous jumps in both the stock and the variance components. It is important to distinguish our paper from theirs: their method is based on a discretization scheme of the partial integro-differential equation (PIDE) formulations, while our method is focused on Markov chain approximations and Fourier techniques using characteristic functions. Their method is based on numerical finite difference techniques, while our method is of a probabilistic nature. Our method can handle a general class of stochastic volatility models with general correlation structures between the two driving Brownian motions.¹ Transform methods have also been considered in Fang and Oosterlee (2011) and Zeng and Kwok (2014) for the Heston model, but require an explicit form for the transition density of variance and the conditional characteristic function of integrated variance, which is not generally available.

In this paper, by blending regime switching models, Markov chain approximation techniques and Fourier methods, we develop a fast and accurate approach for pricing Bermudan and American options and (European) barrier options in general stochastic volatility models with jumps, which we refer to as the PROJ method. This includes discretely monitored contracts as well as continuously monitored options common in the foreign exchange (Forex) market, and encompasses regime switching jump diffusion models as a special case. The models considered include those of Heston, Hull-White, Stein-Stein, the 3/2 model, the 4/2 model, and the α -

¹Introducing jumps in the variance process, as in Salmi et al. (2014), is left as a topic for future research.

Hypergeometric model with any type of jump amplitude distribution in the return process. We believe that the results presented in this paper are new, of interest to academics and practitioners, and can be used as a comparison benchmark in future studies. Our paper makes the following main contributions:

1. We extend the realm of transform-based pricing methods to incorporate barrier, Bermudan and American options under very general regime switching jump diffusion models, which permit different jump distributions in each regime. Capitalizing on the rapid convergence and simplicity of frame projection with the linear basis, we devise an efficient and accurate pricing algorithm.
2. By adopting a Markov chain approximation approach, we are able to reduce the problem of pricing American options in stochastic volatility models with jumps to that of pricing options in a regime switching jump diffusion model. The approach thus provides a *unified* method for pricing American options in many well-known stochastic volatility models with jumps, which include those of Heston, Hull-White, Scott, Stein-Stein, α -Hypergeometric model, the 3/2 model, and the 4/2 model with virtually any jump amplitude distribution in the asset price.
3. We provide numerical results for American options and barrier options under regime switching jump diffusion models and under stochastic volatility models with jumps. For some of these models, prices have never been reported in the literature. We verify the accuracy of our method against benchmarks based on tree methods, and cases for which prices exist in the literature. Due to the accuracy of our method, these results can be used as reference benchmark values for future research.

The rest of the paper is structured as follows: Section 2 casts the probability settings, introduces the regime switching jump diffusion framework and the weak approximation scheme to stochastic volatility models with jumps using continuous time Markov chains (CTMCs). Section 3 discusses the pricing algorithm for early-exercise contracts. Section 4 illustrates the accuracy and efficiency of the method through numerical examples with Heston model, while Section 5 considers additional stochastic volatility models. Section 6 concludes the paper with future research directions.

2 Regime switching framework

This section introduces the main mathematical framework for pricing European and exotic options under regime switching jump diffusions as well as stochastic volatility models with jumps. Consider a complete probability space $(\Omega, \mathcal{F}, \mathbb{Q})$ under which all processes are defined,

where $\mathbb{Q}$ is the risk-neutral probability measure. We assume the existence of a finite state continuous-time process, denoted by $\alpha(t)$, which describes the current state of the economy at time t , and resides in the set of possible states $\mathcal{M} := \{1, 2, \dots, m_0\}$. The transition dynamics of $\alpha(t)$ are described by the generator matrix $Q = [q_{ij}]_{m_0 \times m_0}$, whose elements q_{ij} satisfy (i) $q_{ii} \leq 0$, and $q_{ij} \geq 0$, if $i \neq j$, and (ii) $\sum_j q_{ij} = 0$, $\forall i \in \mathcal{M}$. In terms of q_{ij} , the current regime makes transitions according to

$$\text{Prob}(\alpha(t + \Delta_t) = j | \alpha(t) = i, \alpha(t'), 0 \leq t' \leq t) = q_{ij} \Delta_t + o(\Delta_t), \quad \forall j \neq i. \quad (1)$$

Let B_t be a standard Brownian motion independent of $\alpha(t)$. The underlying asset price S_t evolves under the risk-neutral measure as

$$\frac{dS_t}{S_{t-}} = (r - q - \lambda_{\alpha(t)} \kappa_{\alpha(t)}) dt + \sigma_{\alpha(t)} dB_t + \int_E [e^{\gamma(y, \alpha(t-))} - 1] \pi^\alpha(dy, dt), \quad (2)$$

where $\kappa_{\alpha(t)} = \int_E [e^{\gamma(y, \alpha(t-))} - 1] \nu_{\alpha(t)}(y) dy$ is finite for each regime, with jump intensity $\lambda(\alpha(t), dy) = \lambda_{\alpha(t)} \nu_{\alpha(t)}(y) dy$, and $\pi^\alpha(dy, dt)$ is the Lévy measure. Here r is the common risk-free interest rate, and $q \geq 0$ the continuous dividend yield.

2.1 Characteristic functions

The methods developed in this paper rely on the characteristic function of the log return process $X_t = \log(S_t/S_0)$, $t \in [0, T]$. For a regime switching (RS) jump diffusion, X_t satisfies

$$X_t = \int_0^t \zeta_{\alpha(t)} dt + \int_0^t \sigma_{\alpha(t)} dB(t) + \int_0^t \int_E \gamma(y, \alpha(t-)) \pi^\alpha(dy, dt), \quad (3)$$

where $\zeta_{\alpha(t)} = r - q - \frac{1}{2} \sigma_{\alpha(t)}^2 - \lambda_{\alpha(t)} \kappa_{\alpha(t)}$ is the risk-neutral (convexity-adjusted) drift. Each state $j = 1, \dots, m_0$ is characterized by a particular model of dynamics in that state, or equivalently by its characteristic function. Given a time increment of size $\Delta_t > 0$, and $\xi \in \mathbb{R}$, define for each state $j = 1, \dots, m_0$

$$\tilde{\phi}_{X_{\Delta_t}}^j(\xi) := \mathbb{E}[e^{i\xi X_{\Delta_t}} | \alpha(0 \leq s \leq \Delta_t) = j] := \exp(\psi_j(\xi) \Delta_t), \quad (4)$$

where $\tilde{\phi}_{X_{\Delta_t}}^j(\xi)$ is the risk-neutral characteristic function (chf) of a process which spends the entirety of Δ_t in state j , and $\psi_j(\xi)$ is its *Lévy symbol*. A full model of the process is specified by describing each $\psi_j(\xi)$ along with the generator Q . The characteristic function of a regime switching jump diffusion is characterized by the set

$$\psi_j(\xi) = i\xi \zeta_j - \frac{1}{2} \xi^2 \sigma_j^2 + \lambda_j (\phi_j(\xi) - 1), \quad j = 1, \dots, m_0, \quad (5)$$

where $\phi_j(\xi) = \mathbb{E}[e^{i\xi \gamma(y, \alpha(t))}]$ is the characteristic function of the jump magnitude in state j . In terms of $\phi_j(\xi)$,

$$\zeta_j = r - q - \frac{1}{2} \sigma_j^2 - \lambda_j (\phi_j(-i) - 1). \quad (6)$$

In general, if we are provided with a Lévy symbol $\bar{\psi}_j(\xi)$, then the risk-neutral model is obtained by using $\psi_j(\xi) := r - q - \bar{\psi}_j(-i) + \bar{\psi}_j(\xi)$, so $\zeta_j = r - q - \bar{\psi}_j(-i)$. Because our method incorporates jumps by using only the characteristic function of the jump component, it can be applied with general Lévy processes, including regime switching Lévy models, and stochastic volatility with a Lévy jump component. Next, let's recall a useful result of Buffington and Elliot (2002), which plays a vital role later.

Corollary 1 (*Buffington and Elliot (2002)*) For $\Delta_t > 0$, the characteristic function of X_{Δ_t} is given by the matrix exponential

$$\mathbb{E}[\exp(iX_{\Delta_t}\xi)|\alpha(0)] = \mathbf{1}' \cdot \exp(\Delta_t(Q' + \text{diag}(\psi_1(\xi), \dots, \psi_{m_0}(\xi)))\mathbb{I}(0)), \quad (7)$$

where $\mathbf{1} \in \mathbb{R}^{m_0}$ is a unit vector, and $\mathbb{I}(0) \in \mathbb{R}^{m_0}$ is a vector of zeros, except for the value 1 in the position $\alpha(0)$.

2.2 Stochastic volatility

While regime switching models are an interesting class on their own, the main contribution of this paper is to provide a framework for pricing exotic options under general stochastic volatility models (with jumps). Consider the following two-dimensional system of correlated time-homogeneous diffusions with independent jumps in the asset dynamics

$$\begin{cases} dS_t = S_t\gamma(v_t)dt + m(v_t)S_t dW_t^1 + S_t(e^{J_t} - 1)dN_t, \\ dv_t = \hat{\mu}(v_t)dt + \hat{\sigma}(v_t)dW_t^2, \end{cases} \quad (8)$$

where W_t^1, W_t^2 are Brownian motions with $\mathbb{E}[dW_t^1 dW_t^2] = \rho dt$ for $\rho \in (-1, 1)$. Here N_t is a Poisson process,² and J_t is the jump amplitude, and they are independent of W_t^1 and W_t^2 . In addition, we assume that there exists a constant $C > 0$ such that for all v_1, v_2 in the state space of v_t

$$|\hat{\mu}(v_1) - \hat{\mu}(v_2)| + |\hat{\sigma}(v_1) - \hat{\sigma}(v_2)| \leq C|v_1 - v_2|, \quad (\hat{\mu}(v_1))^2 + (\hat{\sigma}(v_1))^2 \leq C^2(1 + v_1^2). \quad (9)$$

The above conditions guarantee that there exists a unique solution v_t possessing the strong Markov property (see Gihman and Skorohod (1979)). Moreover, we assume that $\hat{\sigma}(\cdot)$ and $m(\cdot)$ are continuously differentiable, with $\hat{\sigma}(\cdot) > 0$ on the domain of v_t .

Let S_t^c denote the continuous part of the stock price, then from the Ito's formula for jump processes, we have

$$\begin{aligned} d\log(S_t) &= \frac{1}{S_t} dS_t^c - \frac{1}{2S_t^2} (dS_t^c)^2 + d \sum_{0 < s \leq t} [\log(S_s) - \log(S_{s-})] \\ &= \left(\gamma(v_t) - \frac{1}{2}m^2(v_t) \right) dt + m(v_t)dW_t^{(1)} + d \sum_{0 < s \leq t} [\log(S_s) - \log(S_{s-})]. \end{aligned} \quad (10)$$

²As mentioned in section 2.1, the method applies more generally to a stochastic volatility model with a Lévy jump component. The special case of a stochastic volatility jump diffusion is considered here for simplicity.

Next, define $\hat{f}(x) := \int_c^x \frac{m(u)}{\hat{\sigma}(u)} du$ with c being a constant, and $h(x) := \mathcal{L}(\hat{f}(x)) = \hat{\mu}(x)\hat{f}'(x) + \frac{1}{2}\hat{\sigma}^2(x)\hat{f}''(x)$. Define $f(v_t, v_0) := \rho(\hat{f}(v_t) - \hat{f}(v_0))$, then we have

$$df(v_t, v_0) = \rho d\hat{f}(v_t) = \rho h(v_t)dt + \rho m(v_t)dW_t^2. \quad (11)$$

Finally, define $W_t^* := \frac{W_t^1 - \rho W_t^2}{\sqrt{1-\rho^2}}$, then one can easily verify that W_t^* is a standard Brownian motion and $\mathbb{E}[dW_t^* dW_t^2] = 0$, from which the two Brownian motions W_t^* and W_t^2 are independent. Next, we plug (11) into (10), and obtain

$$\begin{aligned} d\log(S_t) &= \left(\gamma(v_t) - \frac{1}{2}m^2(v_t) \right) dt + m(v_t)(\rho dW_t^2 + \sqrt{1-\rho^2}dW_t^*) + d \sum_{0 < s \leq t} [\log(S_s) - \log(S_{s-})] \\ &= \left(\gamma(v_t) - \frac{1}{2}m^2(v_t) \right) dt + df(v_t, v_0) - \rho h(v_t)dt + \sqrt{1-\rho^2}m(v_t)dW_t^* \\ &\quad + d \sum_{0 < s \leq t} [\log(S_s) - \log(S_{s-})]. \end{aligned} \quad (12)$$

Let $\tilde{X}_t := \log(S_t/S_0) - f(v_t, v_0)$, then we can rewrite (8) as

$$\begin{cases} d\tilde{X}_t = \left(\gamma(v_t) - \frac{1}{2}m^2(v_t) - \rho h(v_t) \right) dt + \sqrt{1-\rho^2}m(v_t)dW_t^* + d \sum_{0 < s \leq t} [\log(S_s) - \log(S_{s-})], \\ dv_t = \hat{\mu}(v_t)dt + \hat{\sigma}(v_t)dW_t^2. \end{cases} \quad (13)$$

Our approach is as follows: we first approximate v_t by a m_0 -state Markov chain with variance states $\mathbf{v} = \{v_1, v_2, \dots, v_{m_0}\}$ and with the corresponding generator Q governing the transition between variance states (see Section 2.3 for details). We then specify the function $f(\cdot)$ as above such that

$$\log(S_t/S_0) = \tilde{X}_t + f(v_t, v_0), \quad (14)$$

and the dynamics are approximated by the regime switching jump diffusion process

$$d\tilde{X}_t = \mu_{\alpha(t)}dt + \sigma_{\alpha(t)}dW_t^* + d \sum_{0 < s \leq t} [\log(S_s) - \log(S_{s-})], \quad (15)$$

where

$$\begin{cases} \mu_{\alpha(t)} = \gamma(v_{\alpha(t)}) - \frac{1}{2}m^2(v_{\alpha(t)}) - \rho h(v_{\alpha(t)}), \\ \sigma_{\alpha(t)} = \sqrt{1-\rho^2}m(v_{\alpha(t)}), \end{cases} \quad (16)$$

and in particular W_t^* is a standard Brownian motion independent of W_t^2 that drives the variance process. The states $\{v_1, \dots, v_{m_0}\}$ are represented by the chain $\alpha(t)$ on $j = 1, \dots, m_0$. Hence, $\tilde{X}_t$ is of the form of equation (3), and the pricing problem in a stochastic volatility model is reduced to an equivalent pricing problem in a regime switching jump diffusion model.

Once $\tilde{X}_t$ has been identified, the set of chfs describing the dynamics of $\tilde{X}_t$ are

$$\psi_j(\xi) = i\xi\mu_j - \frac{1}{2}\xi^2\sigma_j^2 + \lambda(\phi(\xi) - 1), \quad j = 1, \dots, m_0, \quad (17)$$

Heston	$f(v_k, v_j)$ μ_j σ_j	$\frac{\rho}{\sigma_v}(v_k - v_j)$ $(\frac{\rho\eta}{\sigma_v} - \frac{1}{2})v_j + (r - q - \frac{\rho\eta\theta}{\sigma_v})$ $\sqrt{(1 - \rho^2)}v_j$
4/2	$f(v_k, v_j)$ μ_j σ_j	$\frac{\rho}{\sigma_v}(a(v_k - v_j) + b(\log(v_k) - \log(v_j)))$ $(\frac{a\rho\eta}{\sigma_v} - \frac{a^2}{2})v_j + (\frac{\rho b\sigma_v - b^2}{2} - \frac{b\rho\eta\theta}{\sigma_v})\frac{1}{v_j} + (\frac{\rho\eta}{\sigma_v}(b - a\theta) + r - q - \lambda\kappa - ab)$ $(a\sqrt{v_j} + \frac{b}{\sqrt{v_j}})\sqrt{(1 - \rho^2)}$
Stein-Stein	$f(v_k, v_j)$ μ_j σ_j	$\frac{1}{2}\frac{\rho}{\sigma_v}(v_k^2 - v_j^2)$ $(\frac{\rho\eta}{\sigma_v} - \frac{1}{2})v_j^2 - \frac{\rho\eta\theta}{\sigma_v}v_j + (r - q - \lambda\kappa - \frac{\rho\sigma_v}{2})$ $v_j\sqrt{(1 - \rho^2)}$
3/2	$f(v_k, v_j)$ μ_j σ_j	$\frac{\rho}{\sigma_v}(\log(\hat{v}_k) - \log(\hat{v}_j))$ $\frac{1}{2}(\rho\hat{\sigma}_v - 2\frac{\rho\hat{\eta}\hat{\theta}}{\hat{\sigma}_v} - 1)\frac{1}{\hat{v}_j} + (\frac{\rho\hat{\eta}}{\hat{\sigma}_v} + r - q - \lambda\kappa)$ $\sqrt{\frac{(1 - \rho^2)}{\hat{v}_j}}$
Hull-White	$f(v_k, v_j)$ μ_j σ_j	$\frac{2\rho}{\sigma_v}(\sqrt{v_k} - \sqrt{v_j})$ $(\frac{\rho\sigma_v}{4} - \frac{a_v\rho}{\sigma_v})\sqrt{v_j} - \frac{1}{2}v_j + (r - q - \lambda\kappa)$ $\sqrt{(1 - \rho^2)}v_j$
Scott	$f(v_k, v_j)$ μ_j σ_j	$\frac{\rho}{\sigma_v}(e^{v_k} - e^{v_j})$ $\rho(\frac{\eta}{\sigma_v}(v_j - \theta) - \frac{\sigma_v}{2})e^{v_j} - \frac{e^{2v_j}}{2} + (r - q - \lambda\kappa)$ $e^{v_j}\sqrt{(1 - \rho^2)}$
α -H	$f(v_k, v_j)$ μ_j σ_j	$\frac{\rho}{\sigma_v}(e^{v_k} - e^{v_j})$ $\frac{\rho\theta}{\sigma_v}e^{(1+\alpha_v)v_j} - \rho(\frac{\eta}{\sigma_v} + \frac{\sigma_v}{2})e^{v_j} - \frac{e^{2v_j}}{2} + (r - q - \lambda\kappa)$ $e^{v_j}\sqrt{(1 - \rho^2)}$

Table 1: Transform for pricing with common stochastic volatility models with jumps.

where $\lambda(\phi(\xi) - 1)$ is the state-independent characteristic exponent of the jump component. The necessary transforms as well as the drift and volatility functions $\mu_{\alpha(t)}$ and $\sigma_{\alpha(t)}$ for equation (15) are provided in Table 1 for many popular stochastic volatility models, which we have augmented with jumps.

For example, consider Heston stochastic volatility model with jumps:

$$\mathbf{Hes:} \begin{cases} dS_t = S_t(r - \lambda\kappa)dt + S_t\sqrt{v_t}dW_t^1 + S_t(e^{J_t} - 1)dN_t, \\ dv_t = \eta(\theta - v_t)dt + \sigma_v\sqrt{v_t}dW_t^2, \end{cases} \quad (18)$$

where η is the mean reversion rate, θ is the equilibrium level, and $\sigma_v > 0$ is the volatility of volatility. If we apply the change of variables

$$\tilde{X}_t = \log\left(\frac{S_t}{S_0}\right) - \rho \int_{v_0}^{v_t} \frac{m(u)}{\hat{\sigma}(u)} du = \log\left(\frac{S_t}{S_0}\right) - \frac{\rho}{\sigma_v}(v_t - v_0), \quad (19)$$

with $f(v_t, v_0) = \frac{\rho}{\sigma_v}(v_t - v_0)$, then it follows that

$$\begin{cases} d\tilde{X}_t = \left[(\frac{\rho\eta}{\sigma_v} - \frac{1}{2})v_t + (r - \frac{\rho\eta\theta}{\sigma_v} - \lambda\kappa) \right] dt + \sqrt{(1 - \rho^2)}v_t dW_t^* + d\left[\sum_{k=1}^{N_t} Y_k \right], \\ dv_t = \eta(\theta - v_t)dt + \sigma_v\sqrt{v_t}dW_t^2. \end{cases} \quad (20)$$

Remark 1 An important property of this weak approximation approach is that the characteristic function of the log stock price in the underlying stochastic volatility model is not even required to employ a transform based approach, thus our method can be applied to a broader class of stochastic volatility models where the characteristic functions either do not exist, or

have not been determined in the literature. This distinguishes our approach from the transform-based analysis in the literature, e.g. Duffie et al. (2000). We also do not need the transition density of the variance process to carry out our approach.

2.3 Markov chain approximation of variance

Suppose that under $\mathbb{Q}$ the variance process follows a general diffusion process

$$\begin{cases} dv_t = \hat{\mu}(v_t)dt + \hat{\sigma}(v_t)dW_t^2, \\ v_0 \in \mathbb{R}. \end{cases} \quad (21)$$

Consider a m_0 -state Markov chain on $\mathbf{v} = \{v_1, v_2, \dots, v_{m_0}\}$ with $v_{i-1} < v_i$ and $\mathbf{k} = \{k_1, k_2, \dots, k_{m_0-1}\}$, and we denote the set of grid spacings with $k_i = v_i - v_{i-1}$. Once v_1 and v_{m_0} have been determined (as illustrated in the following Remark 2), our approach is to first define a non-uniform grid as in Tavella and Randall (2000):

$$v_j = v_0 + \bar{\alpha} \sinh \left(c_2 \frac{j}{m_0} + c_1 \left(1 - \frac{j}{m_0} \right) \right), \quad j = 2, \dots, m_0 - 1, \quad (22)$$

where

$$c_1 = \operatorname{arcsinh} \left(\frac{v_1 - v_0}{\bar{\alpha}} \right), \quad c_2 = \operatorname{arcsinh} \left(\frac{v_{m_0} - v_0}{\bar{\alpha}} \right) \quad (23)$$

for $\bar{\alpha} < (v_{m_0} - v_1)$, which we set to $\bar{\alpha} = (v_{m_0} - v_1)/2$.

Next we parameterize the generator Q (as in Lo and Skindilias (2014) for a general diffusion) by

$$q_{ij} = \begin{cases} \frac{\hat{\mu}^-(v_i)}{k_{i-1}} + \frac{\hat{\sigma}^2(v_i) - (k_{i-1}\hat{\mu}^-(v_i) + k_i\hat{\mu}^+(v_i))}{k_{i-1}(k_{i-1} + k_i)}, & \text{if } j = i - 1, \\ \frac{\hat{\mu}^+(v_i)}{k_i} + \frac{\hat{\sigma}^2(v_i) - (k_{i-1}\hat{\mu}^-(v_i) + k_i\hat{\mu}^+(v_i))}{k_i(k_{i-1} + k_i)}, & \text{if } j = i + 1, \\ -q_{i,i-1} - q_{i,i+1}, & \text{if } j = i, \\ 0, & \text{if } j \neq i - 1, i, i + 1, \end{cases} \quad (24)$$

where $a^\pm = \max\{\pm a, 0\}$. Note that if $\mathbf{k}$ is chosen such that

$$0 < \max_{1 \leq i \leq m_0-1} k_i \leq \min_{1 \leq i \leq m_0} \frac{\hat{\sigma}^2(v_i)}{|\hat{\mu}(v_i)|}, \quad (25)$$

then we have

$$\hat{\sigma}^2(v_i) \geq k_{i-1}\hat{\mu}^-(v_i) + k_i\hat{\mu}^+(v_i), \quad (26)$$

so the set of q_{ij} is well defined. That is, $q_{ij} \geq 0, \forall 1 \leq i \neq j \leq m$, and $\sum_{j=1}^{m_0} q_{ij} = 0, i = 1, \dots, m_0$. Moreover, the first two instantaneous moments of the variance process are locally matched to those of the Markov chain:

$$\mathbb{E}[v_{t+\Delta t} - v_t] = \hat{\mu}(v_i)\Delta t, \quad \mathbb{E}[v_{t+\Delta t} - v_t]^2 = (\hat{\sigma}(v_i))^2\Delta t. \quad (27)$$

Therefore the Markov chain satisfies the local consistency conditions, hence will converge weakly to its continuous counterpart (see Kushner (1990)).

Remark 2 To determine the grid boundaries v_1 and v_{m_0} , we fix $t = T/2$ and center the grid about the mean of the variance process v_t by: $v_1 := \max\{\bar{v}, \bar{\mu}(t) - \gamma\bar{\sigma}(t)\}$ and $v_{m_0} := \bar{\mu}(t) + \gamma\bar{\sigma}(t)$. Here we have defined $\bar{\mu}(t) = \mathbb{E}[v_t|v_0]$ and $\bar{\sigma}(t)$ is the standard deviation conditional on v_0 . Further, we define the constants $\gamma = 3 \sim 4$ and $\bar{v} = 0.0001$ for models which require a positive level of v_t .³ For Stein-Stein model, $\bar{v} = 0.01$ is recommended. For Scott model and the α -Hypergeometric model, v_t is allowed to become negative, so we take $\bar{v} = -\infty$.

Remark 3 We note that for Heston model, 3/2 model, and 4/2 model (considered in Section 3.1),

$$\bar{\mu}(t) = e^{-\eta t} v_0 + \theta(1 - e^{-\eta t}), \quad \bar{\sigma}^2(t) = \frac{\sigma_v^2}{\eta} v_0 (e^{-\eta t} - e^{-2\eta t}) + \frac{\theta \sigma_v^2}{2\eta} (1 - e^{-\eta t} + e^{-2\eta t}).$$

In the Stein-Stein and Scott models, we have

$$\bar{\mu}(t) = e^{-\eta t} v_0 + \theta(1 - e^{-\eta t}), \quad \bar{\sigma}^2(t) = \frac{\sigma_v^2}{2\eta} (1 - e^{-2\eta t}).$$

For Hull-White model, we have

$$\bar{\mu}(t) = v_0 e^{a_v t}, \quad \bar{\sigma}^2(t) = v_0^2 e^{2a_v t} (e^{\sigma_v^2 t} - 1).$$

2.4 Biorthogonal Projection

In order to price contracts in regime switching jump diffusion and stochastic volatility models, we employ the option pricing method of frame projection introduced in Kirkby (2015) that orthogonally projects the probability density function of a random variable onto a Riesz basis, which is a special class of frame. Let X be a random variable on $\mathbb{R}$ with unknown density f_X . Given a compactly supported symmetric function $\varphi(\nu)$ called the *generator*, a *resolution* $a > 0$, and a reference point ν_1 , the infinite dimensional space $\mathcal{M}_a := \overline{\text{span}}\{\varphi_{a,n}\}_{n \in \mathbb{Z}}$ is formed from the basis elements $\varphi_{a,n}(\nu) := a^{1/2} \varphi(a(\nu - \nu_n))$ centered over $\nu_n = \nu_1 + (n - 1)/a$. The orthogonal projection $P_{\mathcal{M}_a} f_X$ of f_X onto $\mathcal{M}_a$ is determined by its *dual basis* $\{\tilde{\varphi}_{a,n}\}_{n \in \mathbb{Z}}$:

$$P_{\mathcal{M}_a} f_X(\nu) = \sum_{n \in \mathbb{Z}} \langle f_X, \tilde{\varphi}_{a,n} \rangle \varphi_{a,n}(\nu),$$

where the dual basis is *biorthogonal*, that is $\langle \varphi_{a,k}, \tilde{\varphi}_{a,n} \rangle = \mathbb{1}_{\{k=n\}}$ (orthogonal bases are a special case). The dual basis resides in the same space as $\varphi_{a,n}$, and can be characterized by the Fourier transform of the generator, $\widehat{\tilde{\varphi}}$, which for the linear basis is given by

$$\widehat{\tilde{\varphi}}(\xi) = \frac{12 \sin^2(\xi/2)}{\xi^2(2 + \cos(\xi))}, \quad (28)$$

³If moments of the variance process are unknown, the grid can be fixed using $v_1 = \beta_1 v_0$ and $v_{m_0} = \beta_2 v_0$. For example, $\beta_1 = 10^{-3}$ and $\beta_2 = 4$.

and can be derived for higher order B-spline bases (see Kirkby (2015) for further details, and Kirkby (2016) for an application of higher order B-splines to Asian options). The projection coefficients satisfy a closed-form integral

$$\langle f_X, \tilde{\varphi}_{a,n} \rangle = \frac{a^{-1/2}}{\pi} \Re \left[\int_0^\infty \exp(-i\nu_n \xi) \cdot \phi_X(\xi) \widehat{\tilde{\varphi}}\left(\frac{\xi}{a}\right) d\xi \right], \quad (29)$$

assuming the characteristic function $\phi_X(\xi)$ is known. In this work, we utilize the linear B-spline generator $\varphi(\nu) = \varphi^{[1]}(\nu)$ defined by

$$\varphi^{[1]}(\nu) = (1 + \nu) \cdot \mathbb{1}_{[-1,0]}(\nu) + (1 - \nu) \cdot \mathbb{1}_{[0,1]}(\nu). \quad (30)$$

First we restrict $\mathcal{M}_a$ to the finite set $\{\nu_n\}_{n=1}^N$, where $\nu_n = (1 - N/2)\Delta + (n - 1)/a$, where the choice of a and N will be discussed below. If we define an N -point frequency grid to satisfy the Nyquist requirement

$$\Delta_\xi = 2\pi a/N, \quad \xi_n = (n - 1)\Delta_\xi, \quad n = 1, \dots, N, \quad (31)$$

and constant $\Upsilon_{a,N} := 32a^4/N$, the coefficients $a^{1/2}\Upsilon_{a,N} \cdot \bar{\beta}_{a,n} \approx \beta_{a,n} = \langle f_X, \tilde{\varphi}_{a,n} \rangle$ are found using the exponentially convergent discretization of (29)

$$\{\bar{\beta}_{a,n}\}_{n=1}^N := \Re \{ \mathcal{D} \{ H_j \} \}, \quad \mathcal{D}_n \{ H_j \} = \sum_{j=1}^N e^{-i\frac{2\pi}{N}(j-1)(n-1)} H_j, \quad n = 1, \dots, N, \quad (32)$$

where $\mathcal{D}$ is the discrete Fourier transform (DFT). From (29) and the dual transform in (28), the DFT input vector $\{H_j\}_{j=1}^N$ is defined by

$$H_1 = \frac{1}{24a^2}, \quad H_n = \exp(-i\nu_1 \xi_n) \cdot \phi_X(\xi_n) \cdot \frac{(\sin(\xi_n/(2a))/\xi_n)^2}{2 + \cos(\xi_n/a)}, \quad n \geq 2, \quad (33)$$

where $\nu_1 := (1 - N/2)\Delta$ centers the expansion around zero. Note that we have chosen the linear basis in this work for its simplicity. However, higher order bases are likely to improve accuracy, at the cost of a more complicated algorithm.

Remark 4 *In light of Corollary 1, the PROJ method of Kirkby (2015) can be directly applied to price European call options with arbitrary payoffs $g(S_{\Delta_t})$ where the underlying model is a regime switching Lévy process as in Section 2. Moreover, the Markov chain approximation of Section 2.3 provides an alternative approach to price European options under stochastic volatility models when the characteristic function of $\ln(S_t/S_0)$ is unavailable or ill-behaved.*

2.5 Barrier options

Without loss of generality⁴ we consider knock-out barrier options, which are contracts that become worthless if the underlying enters a prescribed knock-out region. Let $\mathcal{C}$ denote the

⁴Since knock-in options can be priced by parity.

continuation region, and $\mathcal{C}^c$ the knock-out region for X_t where $X_t = \ln(S_t/S_0)$. For a double barrier option with knockout barriers L and U in asset space, $\mathcal{C} = [l_x, u_x]$ where $l_x := \ln(L/S_0)$ and $u_x := \ln(U/S_0)$ in log space. Define the terminal payoff $G_{S_0}(X) = g(S_0 \exp(X))$ for some $g(\cdot)$, and fix a set of monitoring dates $t_m = m\Delta_t$, $m = 0, \dots, M$, where $\Delta_t = T/M$. Let $\mathcal{V}_m^*(X_{t_m}, \alpha(t_m))$ denote the value of a barrier option at time t_m which has not previously been knocked out, where $X_{t_m} = \ln(S_{t_m}/S_0)$ and $\alpha(t_m) \in \{1, \dots, m_0\}$ is the current state, and represents the current variance level in the case of stochastic volatility. The barrier option price can be found by backward recursion with

$$\begin{cases} \mathcal{V}_M^*(X_M, \alpha_M) = G_{S_0}(X_M) \\ \mathcal{V}_m^*(X_m, \alpha_m) = e^{-r\Delta_t} \mathbb{E} [\mathcal{V}_{m+1}^*(X_{m+1}, \alpha_{m+1}) \mathbb{1}_{\{X_m \in \mathcal{C}\}} | X_m, \alpha_m] \quad m = M-1, \dots, 0, \end{cases} \quad (34)$$

where⁵ $X_m := X_{t_m}$ and $\alpha_m = \alpha(t_m)$. By definition, $\mathcal{V}_{m+1}^*(X_{m+1}, \alpha_{m+1}) = 0$ for $X_{m+1} \in \mathcal{C}^c$. Let $G_{j,k}^m(x) := \{X_{m+1} = x, \alpha_m = j, \alpha_{m+1} = k\}$, and define the state transition probabilities

$$P_{jk}^{\Delta_t} = \mathbb{Q}[\alpha(t + \Delta_t) = k | \alpha(t) = j], \quad j, k = 1, \dots, m_0.$$

Then with $\alpha_m = j$ and $X_m = x \in \mathcal{C}$, we have

$$\begin{aligned} \mathcal{V}_m^*(x, j) &= e^{-r\Delta_t} \mathbb{E} [\mathcal{V}_{m+1}^*(X_{m+1}, \alpha_{m+1}) | X_m = x, \alpha_m = j] \\ &= e^{-r\Delta_t} \sum_{k=1, \dots, m_0} P_{j,k}^{\Delta_t} \mathbb{E} [\mathcal{V}_{m+1}^*(X_{m+1}, k) | G_{j,k}^m(x)] \\ &= e^{-r\Delta_t} \sum_{k=1, \dots, m_0} P_{j,k}^{\Delta_t} \int_{\mathcal{C}} \mathcal{V}_{m+1}^*(y, k) p_{j,k}(y|x) dy, \end{aligned}$$

where we have defined the set of transition densities for the log return process

$$p_{j,k}(y|x) = \mathbb{Q}[X(\Delta_t) \in y + dy | X(0) = x, \alpha(0) = j, \alpha(\Delta_t) = k], \quad j, k = 1, \dots, m_0.$$

For a regime switching model, the set of transition densities can be recovered from

$$\mathbb{E}[e^{i\xi X_{\Delta_t}} | G_{j,k}] = \frac{1}{P_{j,k}^{\Delta_t}} \mathcal{E}_{k,j}(\xi), \quad \mathcal{E}(\xi) = \exp(\Delta_t (Q' + \text{diag}(\psi_1(\xi), \dots, \psi_{m_0}(\xi))).$$

For a stochastic volatility model with log return process given by $X_{\Delta_t} = \tilde{X}_{\Delta_t} + f(v_{\Delta_t}, v_0)$, we have $\mathbb{E}[e^{i\xi X_{\Delta_t}} | G_{j,k}] = \mathcal{E}_{k,j}(\xi) \cdot \exp(i\xi f(v_k, v_j)) / P_{j,k}^{\Delta_t}$, which leads us to define

$$\tilde{\mathcal{E}}_{k,j}(\xi) := \begin{cases} \mathcal{E}_{k,j}(\xi), & \text{regime-switching model} \\ \mathcal{E}_{k,j}(\xi) \cdot \exp(i\xi f(v_k, v_j)), & \text{stochastic volatility model.} \end{cases} \quad (35)$$

This allows a unified treatment of regime-switching and stochastic volatility models in the pricing algorithms that follow. Once the generator Q has been determined⁶, the different

⁵A European option can be priced recursively by setting $\mathcal{C} = (-\infty, \infty)$.

⁶In the regime-switching case Q is provided as a model parameter, while for a stochastic volatility model it is determined as in section 2.3.

definitions of $\tilde{\mathcal{E}}_{k,j}(\xi)$ are the main difference between SV and RS models in the algorithm. Noting that $p_{j,k}(y|x) = p_{j,k}(\nu)$ where $\nu = y - x$, we have

$$\begin{aligned} p_{j,k}(\nu) &\approx \sum_{n \in \mathbb{Z}} \left(\int_{-\infty}^{\infty} p_{j,k}(\eta) \tilde{\varphi}_{a,n}(\eta) d\eta \right) \varphi_{a,n}(\nu) \\ &= \frac{a^{-1/2}}{\pi} \frac{1}{P_{j,k}^{\Delta t}} \sum_{n \in \mathbb{Z}} \Re \left\{ \int_0^{\infty} \exp(-i\nu_n \xi) \cdot \tilde{\mathcal{E}}_{k,j}(\xi) \cdot \widehat{\varphi}(\xi/a) d\xi \right\} \varphi_{a,n}(\nu). \end{aligned}$$

Hence,

$$P_{j,k}^{\Delta t} \cdot p_{j,k}(\nu) \approx \Upsilon_{a,N} \sum_{1 \leq n \leq N} \bar{\beta}_{a,n}^{j,k} \cdot a^{1/2} \varphi_{a,n}(\nu) := \bar{p}_{j,k}(\nu),$$

where the set of $\{\bar{\beta}_{a,n}^{j,k}\}_{n=1}^N$ is computed⁷ as in Section 2.4, for each $j, k = 1, \dots, m_0$. In this case, the DFT input in equation (33) is defined by setting $\phi_X(\xi_n) = \tilde{\mathcal{E}}_{k,j}(\xi_n)$:

$$H_1^{j,k} = \frac{\tilde{\mathcal{E}}_{k,j}(\xi_1)}{24a^2}, \quad H_n^{j,k} = \exp(-i\nu_1 \xi_n) \cdot \tilde{\mathcal{E}}_{k,j}(\xi_n) \cdot \frac{(\sin(\xi_n/(2a))/\xi_n)^2}{2 + \cos(\xi_n/a)}, \quad n \geq 2. \quad (36)$$

Moreover, the basis elements are centered over a grid $\{\nu_n\}_{n=1}^N$ defined in Section 2.4.

Now define a grid $\{x_n\}_{n=1}^{N/2}$ for the continuation region $\mathcal{C}$ in log asset space as in Section 2.5.1 below. We approximate the value at node $(X_m, \alpha_m) = (x_n, j)$ by replacing the true (unknown) density with its orthogonal projection

$$\begin{aligned} \mathcal{V}_m^*(x_n, j) &= e^{-r\Delta t} \sum_{k=1, \dots, m_0} \int_{\mathcal{C}} \mathcal{V}_{m+1}^*(y, k) \cdot P_{j,k}^{\Delta t} \cdot p_{j,k}(y|x_n) dy \\ &= e^{-r\Delta t} \sum_{k=1, \dots, m_0} \int_{\mathcal{C}_n} \mathcal{V}_{m+1}^*(x_n + \nu, k) \cdot P_{j,k}^{\Delta t} \cdot p_{j,k}(\nu) d\nu \\ &\approx e^{-r\Delta t} \sum_{k=1, \dots, m_0} \int_{\mathcal{C}_n} \mathcal{V}_{m+1}^*(x_n + \nu, k) \cdot \bar{p}_{j,k}(\nu) d\nu, \end{aligned} \quad (37)$$

where $\mathcal{C}_n := \{\nu \in \mathbb{R} : x_n + \nu \in \mathcal{C}\}$. For each $k = 1, \dots, m_0$, we can represent the convolution from (38) in the following form which facilitates an efficient Toeplitz-based algorithm:

$$\begin{aligned} \int_{\mathcal{C}_n} \mathcal{V}_{m+1}^*(x_n + \nu, k) \cdot \bar{p}_{j,k}(\nu) d\nu &= \Upsilon_{a,N} \sum_{l=1}^N \bar{\beta}_l^{j,k} \cdot a^{1/2} \int_{\mathcal{C}_n} \mathcal{V}_{m+1}^*(x_n + \nu, k) \varphi_{a,l}(\nu) d\nu \\ &\approx \Upsilon_{a,N} \sum_{l=1}^{N/2} \bar{\beta}_{N/2+(l-n)}^{j,k} \cdot a^{1/2} \int_{\mathcal{C}_n} \mathcal{V}_{m+1}^*(x_n + \nu, k) \varphi_{a,N/2+(l-n)}(\nu) d\nu, \end{aligned} \quad (38)$$

where $\bar{\beta}_l^{j,k} \equiv \bar{\beta}_{a,l}^{j,k}$ are the projection coefficients of $\bar{p}_{j,k}(\nu)$.

Remark 5 The approximation (38) holds exactly for a double barrier option with $\mathcal{C} = [l_x, u_x] = [x_1, x_n]$ (since the value is zero outside of $\mathcal{C}$), but introduces a truncation error in the case of an

⁷As discussed below, the set of $\{\bar{\beta}_{a,n}^{j,k}\}_{n=1}^N$ are not stored explicitly. We instead store a set of vectors $\boldsymbol{\mu}^{j,k}$ defined in equation (39)

unbounded domain, particularly at the boundaries. For example, at the leftmost grid point x_1 , only the terms $\bar{\beta}_{N/2}^{j,k}, \bar{\beta}_{N/2+1}^{j,k}, \dots, \bar{\beta}_{N-1}^{j,k}$ corresponding to the right half of the transition density are present in the summation in (38), as seen in the first row of $\mathbf{T}_{j,k}^{\Delta t}$ below. However, the error is easily controlled by choosing $\hat{\alpha} = (x_n - x_1)/2$ sufficiently large, as in the experiments below. Alternatively, Kirkby (2017b) proposes a grid extension method which approximates the convolution outside of $[x_1, x_n]$ (resulting in accurate approximations for reduced $\hat{\alpha}$), that can be adapted to the present case. To simplify the presentation, we consider the case without boundary modification.

To efficiently determine values from (38) along the full grid, we introduce a set of Toeplitz matrices, where $\Upsilon_{a,N}^{\Delta t} = \exp(-r\Delta t) \cdot \Upsilon_{a,N}$:

$$\mathbf{T}_{j,k}^{\Delta t} = \Upsilon_{a,N}^{\Delta t} \cdot \begin{bmatrix} \bar{\beta}_{N/2}^{j,k} & \bar{\beta}_{N/2+1}^{j,k} & \bar{\beta}_{N/2+2}^{j,k} & \cdots & \bar{\beta}_{N-1}^{j,k} \\ \bar{\beta}_{N/2-1}^{j,k} & \bar{\beta}_{N/2}^{j,k} & \bar{\beta}_{N/2+1}^{j,k} & \cdots & \bar{\beta}_{N-2}^{j,k} \\ \bar{\beta}_{N/2-2}^{j,k} & \bar{\beta}_{N/2-1}^{j,k} & \bar{\beta}_{N/2}^{j,k} & & \bar{\beta}_{N-3}^{j,k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \bar{\beta}_1^{j,k} & \bar{\beta}_2^{j,k} & \bar{\beta}_3^{j,k} & \cdots & \bar{\beta}_{N/2}^{j,k} \end{bmatrix}, \quad j, k = 1, \dots, m_0.$$

In particular, to calculate $\mathbf{T}_{j,k}^{\Delta t} u$, for $u = (u_1, \dots, u_{N/2})^\top$, we define

$$\boldsymbol{\mu}^{j,k} := \mathcal{D} \left\{ \Upsilon_{a,N}^{\Delta t} \cdot (\bar{\beta}_{N/2}^{j,k}, \dots, \bar{\beta}_1^{j,k}, 0, \bar{\beta}_{N-1}^{j,k}, \dots, \bar{\beta}_{N/2+1}^{j,k})^\top \right\}, \quad j, k = 1, \dots, m_0, \quad (39)$$

and $\mathbf{T}_{j,k}^{\Delta t} u$ is given by the first $N/2$ elements of

$$p = \mathcal{D}^{-1} \left\{ \boldsymbol{\mu}^{j,k} \circ \mathcal{D} \left\{ (u_1, \dots, u_{N/2}, 0, \dots, 0)^\top \right\} \right\}, \quad (40)$$

where $v \circ u$ is the element-wise (Hadamard) product of v with u , and the vector of zeros is of length $N/2$. Since a computation of the form $\mathbf{T}_{j,k}^{\Delta t} u$ is required at each time step of the algorithm, we will store the set of $\boldsymbol{\mu}^{j,k}$ at initialization for future use. In particular, $\mathbf{T}_{j,k}^{\Delta t}$ is never explicitly constructed in the algorithm, and $\boldsymbol{\mu}^{j,k}$ is stored instead. The full initialization routine is summarized by Algorithm 1 in Appendix A, which applies to barrier as well as Bermudan options. Note that given the algorithm's structure, a single vector $\{H_n\}_{n=1}^N$ can be used to store $H_n^{j,k}$ at each iteration, and similarly for $\bar{\beta}_n^{j,k}$.

Given the Toeplitz representation $\mathbf{T}_{j,k}^{\Delta t}$ of the set of transition densities with j fixed, the values $\mathcal{V}_m^*(x_n, j)$ in equation (38) are computed for the full set of $\{x_n\}_{n=1}^{N/2}$ with complexity $\mathcal{O}(m_0 \cdot N \log_2(N))$ using

$$\{\mathcal{V}_m(x_n, j)\}_{n=1}^{N/2} := \sum_{k=1, \dots, m_0} \mathbf{T}_{j,k}^{\Delta t} \boldsymbol{\theta}_k^{m+1}, \quad m = M-1, \dots, 0,$$

where for each $m = 1, \dots, M$, $\boldsymbol{\theta}_k^m$ is a vector with⁸

$$\theta_{n,k}^m = a^{1/2} \int_{C_n} \mathcal{V}_m(x_n + \nu, k) \varphi_{a,n}(\nu) d\nu, \quad k = 1, \dots, m_0, \quad n = 1, \dots, N/2,$$

⁸Note that in this equation, $\varphi_{a,n}(\nu)$ is centered over x_n .

where $\mathcal{C}_n := \{\nu \in \mathbb{R} : x_n + \nu \in \mathcal{C}\}$ and $\mathcal{V}_m(x_n, j)$ is used to denote the approximation of the true value, $\mathcal{V}_m^*(x_n, j)$. The set of dual or *theta coefficients*, $\{\theta_{n,j}^m\}_{n=1}^{N/2}$ for each $j = 1, \dots, m_0$ are required at each stage m . We use the following closed-form approximation:

$$\theta_{n,j}^m := \begin{cases} [13\mathcal{V}_m(x_1, j) + 15\mathcal{V}_m(x_2, j) - 5\mathcal{V}_m(x_3, j) + \mathcal{V}_m(x_4, j)] / 48 & n = 1 \\ [\mathcal{V}_m(x_{n-1}, j) + 10\mathcal{V}_m(x_n, j) + \mathcal{V}_m(x_{n+1}, j)] / 12 & n = 2, \dots, N/2 - 1 \\ [13\mathcal{V}_m(x_{N/2}, j) + 15\mathcal{V}_m(x_{N/2-1}, j) - 5\mathcal{V}_m(x_{N/2-2}, j) + \mathcal{V}_m(x_{N/2-3}, j)] / 48 & n = N/2 \end{cases} \quad (41)$$

which follows by a local quadratic approximation of each $\mathcal{V}_m(x_n, j)$ (see (55) for details). As shown in the error analysis, the error introduced by this approximation is $\mathcal{O}(\Delta^3)$, and will not affect the rate of convergence with the linear basis.

Thus, after initialization, in order to find $\mathcal{V}_m(x_n, j)$ for all $j \in \{1, \dots, m_0\}$, it requires $\mathcal{O}(m_0^2 \cdot N \log_2(N))$. The algorithm complexity is thus $\mathcal{O}(M \cdot m_0^2 \cdot N \log_2(N) + N \cdot m_0^3)$, where the final terms derives from populating the matrix exponentials $\tilde{\mathcal{E}}(\xi_n)$ for $n = 1, \dots, N$, each at a cost of $\mathcal{O}(m_0^3)$.

2.5.1 Barrier Grid Selection and Terminal Payoff

The log-asset grid, $\{x_n\}_{n=1}^{N/2}$, is adapted to the continuation region, $\mathcal{C}$, for various contracts. We begin with a choice of $N \in \mathbb{N}_+$, the number of basis elements, and the grid width parameter $\hat{\alpha} > 0$, which determines the truncated density support on $(-\hat{\alpha}, \hat{\alpha})$. For each contract type we initially define

$$\tilde{\Delta} := 2\hat{\alpha}/(N - 1), \quad \tilde{a} := 1/\tilde{\Delta}.$$

For down-and-out options with $\mathcal{C} = [l_x, \infty)$, $x_1 := l_x$ coincides with the knock-out barrier, and the grid $x_n = x_1 + (n - 1)\Delta$, $n = 1, \dots, N/2$ is defined by⁹

$$x_1 := l_x, \quad n_0 := \lfloor 1 - l_x \cdot \tilde{a} \rfloor, \quad \Delta := l_x/(1 - n_0), \quad a := 1/\Delta, \quad (42)$$

which satisfies $x_{n_0} = 0$. Once N and Δ have been fixed, the frequency grid is generated as in equation (31).

For one-sided barrier options our grid choice aligns the barrier and initial price (both in log space) with the grid, while the strike might not belong. Define the nearest grid point left of $\log(K/S_0)$ (strike in log space)

$$\bar{n} := \lfloor (a \cdot (\log(K/S_0) - x_1) + 1) \rfloor. \quad (43)$$

⁹We use a log asset grid of size $N/2$ to employ the Toeplitz-based convolution represented by (39)-(40).

Puts	$\bar{\delta}_j^{put}$	δ_j^{put}
$j = 0$	$\zeta (1 - \frac{1}{2}\zeta)$	$\frac{\zeta}{18} [4(2 - \zeta)e^{\bar{\rho}/2} + 5 \cdot ((1 - \zeta_-)e^{\rho_-} + (1 - \zeta_+)e^{\rho_+})]$
$j = 1$	$\frac{\zeta^2}{2}$	$\frac{\zeta}{18} \cdot e^{-\Delta} \cdot [4\zeta \cdot e^{\bar{\rho}/2} + 5(\zeta_-e^{\rho_-} + \zeta_+e^{\rho_+})]$
Calls	$\bar{\delta}_j^{call}$	δ_j^{call}
$j = 0$	$\frac{1}{2} + \zeta (\frac{\zeta}{2} - 1)$	$e^{(\bar{\rho}+\Delta)/2} \frac{\sigma^2}{18} [5 \cdot ((1 - q_-)e^{\Delta\sigma_-} + (1 - q_+)e^{\Delta\sigma_+}) + 4]$
$j = 1$	$\sigma - \frac{1}{2}\sigma^2$	$e^{(\bar{\rho}-\Delta)/2} \frac{\sigma}{18} \cdot [4(\zeta + 1) + 5((\frac{\zeta+1}{2} + \sigma_-)e^{\Delta\sigma_-} + (\frac{\zeta+1}{2} + \sigma_+)e^{\Delta\sigma_+})]$

Table 2: Terminal payoff coefficients derived from a three point Gaussian quadrature, where $q_{\pm} := (1 \pm \sqrt{3/5})/2$, $\zeta_{\pm} := \zeta \cdot q_{\pm}$, and $\rho_{\pm} := \bar{\rho} \cdot q_{\pm}$.

With the formulae defined in Table 2, we can derive the DOP payoff coefficients :

$$\theta_n^M = \begin{cases} \frac{K}{2} - L \cdot \vartheta_{[0,1]}, & n = 1 \\ K - e^{x_n} \cdot S_0 \cdot (\vartheta_{[-1,0]} + \vartheta_{[0,1]}), & n = 2, \dots, \bar{n} - 1 \\ K \left(\frac{1}{2} + \bar{\delta}_0^{put} - e^{-\bar{\rho}} (\vartheta_{[-1,0]} + \delta_0^{put}) \right), & n = \bar{n} \\ K (\bar{\delta}_1^{put} - e^{\Delta-\bar{\rho}} \cdot \delta_1^{put}), & n = \bar{n} + 1 \\ 0, & n = \bar{n} + 2, \dots, N/2 \end{cases} \quad (44)$$

where $\theta_{n,j}^M \equiv \theta_n^M$ are identical for $j = 1, \dots, m_0$. We have also defined the constants

$$\begin{aligned} \vartheta_{[-1,0]} &:= \frac{e^{-\Delta/2}}{18} \left(4 + 5 \cosh(\sqrt{15}\Delta/10) + \sqrt{15} \sinh(\sqrt{15}\Delta/10) \right) \\ \vartheta_{[0,1]} &:= \frac{e^{\Delta/2}}{18} \left(4 + 5 \cosh(\sqrt{15}\Delta/10) - \sqrt{15} \sinh(\sqrt{15}\Delta/10) \right) \end{aligned} \quad (45)$$

Similarly, we define the DOC coefficients

$$\theta_n^M = \begin{cases} 0 & n \leq \bar{n} - 1 \\ K (\delta_0^{call} e^{-\bar{\rho}} - \bar{\delta}_0^{call}) & n = \bar{n} \\ K (e^{\Delta-\bar{\rho}} (\vartheta_{[0,1]} + \delta_1^{call}) - (1/2 + \bar{\delta}_1^{call})) & n = \bar{n} + 1 \\ S_0 e^{x_n} \cdot (\vartheta_{[-1,0]} + \vartheta_{[0,1]}) - K & n = \bar{n} + 2, \dots, N/2 \end{cases} \quad (46)$$

For up-and-out contracts, double barriers and rebates, terminal coefficients and the grid choice are provided in Kirkby (2017b). Algorithm 2 in Appendix A summarizes the procedure for knock-out barrier options, while knock-in options are easily priced using parity relations.

3 Early exercise contracts

Early-exercise contracts are priced in a similar manner to barrier options, with a few modifications. In general, the Bermudan option price can be found by backward recursion with

$$\begin{cases} \mathcal{V}_M^*(X_M, \alpha_M) = G_{S_0}(X_M) \\ \mathcal{C}_m^*(X_m, \alpha_m) = e^{-r\Delta t} \mathbb{E} [\mathcal{V}_{m+1}^*(X_{m+1}, \alpha_{m+1}) | X_m, \alpha_m] & m = M-1, \dots, 0, \\ \mathcal{V}_m^*(X_m, \alpha_m) = \max\{\mathcal{C}_m^*(X_m, \alpha_m), G_{S_0}(X_m)\}, \end{cases} \quad (47)$$

where $G(X_m)$ is the early exercise payoff at time t_m , and $\mathcal{C}_m^*(X_m, \alpha_m)$ is the continuation value as of t_m . Theoretical values are denoted with a superscript star to distinguish them from the approximations to follow.

Let $\{x_n\}_{n=1}^{N/2}$ define a uniformly spaced grid in log asset space (see Remark 10 for details). For a Bermudan option, the continuation value satisfies

$$\begin{aligned} \mathcal{C}_m^*(x_n, j) &= e^{-r\Delta t} \mathbb{E} [\mathcal{V}_{m+1}^*(X_{m+1}, \alpha_{m+1}) | X_m = x_n, \alpha_m = j] \\ &\approx e^{-r\Delta t} \sum_{k=1, \dots, m_0} \int_{-\infty}^{\infty} \mathcal{V}_{m+1}(x_n + \nu, k) \cdot \bar{p}_{j,k}(\nu) d\nu := \mathcal{C}_m(x_n, j), \end{aligned}$$

which requires the set of *value coefficients*, θ_j^{m+1} , defined at time t_{m+1} by

$$\theta_{n,j}^{m+1} := a^{1/2} \int_{x_{n-1}}^{x_{n+1}} \mathcal{V}_{m+1}(x, j) \varphi_{a,n}(x) dx = a^{1/2} \int_{x_{n-1}}^{x_{n+1}} \max\{\mathcal{C}_{m+1}(x, j), G_{S_0}(x)\} \varphi_{a,n}(x) dx, \quad (48)$$

where we note that here $\varphi_{a,n}(x)$ is centered over the grid point x_n , and is supported on $[x_{n-1}, x_{n+1}]$. The integral is evaluated upon determining the set of early-exercise points, which separate log asset space into a continuation and an exercise region. At time t_m , we define for each $j = 1, \dots, m_0$ the *early-exercise point*¹⁰ $\hat{x}^m(j)$, which satisfies $G_{S_0}(\hat{x}^m(j)) = \mathcal{C}_m(\hat{x}^m(j), j)$. For concreteness, we consider the case of a Bermudan put option, with $G_{S_0}(x) = (K - S_0 \exp(x))^+$ (calls are priced similarly). The point $\hat{x}^m(j)$ is found by locating the bracketing left index and grid point

$$\hat{n}(j) = \max\{1 \leq n \leq \bar{n} : G_{S_0}(x_n) - \mathcal{C}_m(x_n, j) \geq 0\}, \quad x_{\hat{n}(j)} = x_1 + (\hat{n}(j) - 1)\Delta, \quad (49)$$

where $\bar{n}$ is defined in equation (43), so the early exercise point satisfies $x_{\hat{n}(j)} \leq \hat{x}^m(j) < x_{\hat{n}(j)+1}$. Once $\hat{n}(j)$ is found, we use the linear approximation

$$\hat{x}^m(j) \approx x_{\hat{n}(j)} + \Delta \frac{G_{S_0}(x_{\hat{n}(j)}) - \mathcal{C}_m(x_{\hat{n}(j)}, j)}{(G_{S_0}(x_{\hat{n}(j)}) - \mathcal{C}_m(x_{\hat{n}(j)}, j)) - (G_{S_0}(x_{\hat{n}(j)+1}) - \mathcal{C}_m(x_{\hat{n}(j)+1}, j))}, \quad (50)$$

which is found by equating the linear approximations of $G_{S_0}(x)$ and $\mathcal{C}_m(x, j)$, and solving for the point of intersection, $\hat{x}^m(j)$. Starting with the set θ_k^M defined in equation (56), the value

¹⁰These early exercise points are unique because we can think of them as taken from the corresponding continuous early exercise boundary (for the continuous finite-maturity American option). Thus if this continuous early exercise boundary is uniquely defined, then these early exercise points will also be uniquely defined due to the convergence relation between Bermudan options and American options. Note that the stochastic volatility dynamics considered in our paper is a special case of the model (6) and (7) in Detemple and Tian (2002), then from Proposition 2 on page 922 of Detemple and Tian (2002), we have that the continuous early exercise boundary is unique. We would like to thank an anonymous referee for raising this point.

recursion becomes

$$\{\mathcal{C}_m(x_n, j)\}_{n=1}^{N/2} := \sum_{k=1, \dots, m_0} \mathbf{T}_{j,k}^{\Delta t} \boldsymbol{\theta}_k^{m+1}, \quad m = M-1, \dots, 0,$$

where $\boldsymbol{\theta}_k^{m+1}$ is defined in equation (48).

Corollary 2 Suppose for $j \in \{1, \dots, m_0\}$ that we have identified an index $\hat{n}(j)$ such that $x_{\hat{n}(j)} \leq \hat{x}^m(j) < x_{\hat{n}(j)+1}$. The value coefficients for $n = 1, \dots, N/2$ of the Bermudan put can be evaluated by a closed-form three point Gaussian quadrature approximation, where $\mathcal{C}_{n,j} := \mathcal{C}_m(x_n, j)$:

$$\theta_{n,j}^m := \begin{cases} \theta_n^M & n = 1, \dots, \hat{n}(j) - 1 \\ K \left(\frac{1}{2} + \bar{\delta}_0^{put} \right) - S_0 \exp(x_{\hat{n}(j)}) \left(\vartheta_{[-1,0]} + \delta_0^{put} \right) \\ \quad + \gamma_1^m \mathcal{C}_{\hat{n}(j)-1,j} + \gamma_2^m \mathcal{C}_{\hat{n}(j),j} + \gamma_3^m \mathcal{C}_{\hat{n}(j)+1,j} & n = \hat{n}(j) \\ K \bar{\delta}_1^{put} - S_0 \exp(x_{\hat{n}(j)+1}) \delta_1^{put} \\ \quad + \gamma_4^m \mathcal{C}_{\hat{n}(j),j} + \gamma_5^m \mathcal{C}_{\hat{n}(j)+1,j} + \gamma_6^m \mathcal{C}_{\hat{n}(j)+2,j} & n = \hat{n}(j) + 1 \\ \frac{1}{12} (\mathcal{C}_{n-1,j} + 10\mathcal{C}_{n,j} + \mathcal{C}_{n+1,j}) & n = \hat{n}(j) + 2, \dots, N/2 - 1 \\ \frac{1}{48} (13\mathcal{C}_{N/2,j} + 15\mathcal{C}_{N/2-1,j} - 5\mathcal{C}_{N/2-2,j} + \mathcal{C}_{N/2-3,j}) & n = N/2, \end{cases} \quad (51)$$

where in stage m we define $\gamma_l^m = \gamma_l^m(j)$ by

$$\begin{aligned} \gamma_1^m &= -\frac{1}{24} + \frac{\zeta^4}{8} - \frac{\zeta^3}{3} + \frac{\zeta^2}{4}, & \gamma_2^m &= \frac{5}{12} - \frac{\zeta^4}{4} + \frac{\zeta^3}{3} + \frac{\zeta^2}{2} - \zeta, & \gamma_3^m &= \frac{1}{8} + \frac{\zeta^4}{8} - \frac{\zeta^2}{4} \\ \gamma_4^m &= \frac{1}{12} - \frac{\zeta^4}{8} + \frac{1}{2} (\zeta^3 - \zeta^2), & \gamma_5^m &= \frac{5}{6} + \frac{\zeta^4}{4} - \frac{2\zeta^3}{3}, & \gamma_6^m &= \frac{1}{12} - \frac{\zeta^4}{8} + \frac{\zeta^3}{6} \end{aligned} \quad (52)$$

and (with time superscripts and dependence on j suppressed to ease notation)

$$\bar{\rho} := \hat{x}^m(j) - x_{\hat{n}(j)}, \quad \zeta := a\bar{\rho}. \quad (53)$$

We have also defined in Table 2 a set of coefficients $\delta_i^{put} = \delta_i^{put}(\zeta, \bar{\rho})$ and $\bar{\delta}_i^{put} = \bar{\delta}_i^{put}(\bar{\rho})$.

Proof: From the definition of $\hat{x}^m(j)$ and $\theta_{n,j}^m$, it follows that

$$\theta_{n,j}^m = a^{1/2} \begin{cases} \int_{x_{n-1}}^{x_{n+1}} G_{S_0}(x) \varphi_{a,n}(x) dx, & n \leq \hat{n}(j) - 1 \\ \int_{x_{n-1}}^{\hat{x}^m(j)} G_{S_0}(x) \varphi_{a,n}(x) dx + \int_{\hat{x}^m(j)}^{x_{n+1}} \mathcal{C}_m(x, j) \varphi_{a,n}(x) dx, & n = \hat{n}(j), \hat{n}(j) + 1 \\ \int_{x_{n-1}}^{x_{n+1}} \mathcal{C}_m(x, j) \varphi_{a,n}(x) dx & n \geq \hat{n}(j) + 2, \end{cases} \quad (54)$$

where $\theta_{n,j}^m = \theta_{n,j}^M \equiv \theta_n^M$ for $n \leq \hat{n}(j) - 1$. We apply a local quadratic interpolation of the value function to evaluate $\theta_{n,j}^m$ for $n = 1, \dots, \hat{n}(j) - 1$. For each interval $[x_{n-1}, x_{n+1}]$, $n = \hat{n}(j), \dots, N/2 - 1$, define

$$\begin{aligned} \tilde{\mathcal{C}}_m(x, j) = & \mathcal{C}_{n-1,j} \frac{(x - x_n)(x - x_{n+1})}{2\Delta^2} - \mathcal{C}_{n,j} \frac{(x - x_{n-1})(x - x_{n+1})}{\Delta^2} \\ & + \mathcal{C}_{n+1,j} \frac{(x - x_{n-1})(x - x_n)}{2\Delta^2}, \end{aligned} \quad (55)$$

and integrate exactly. Given that $\varphi_{a,n}(y)\tilde{\mathcal{C}}_m(y, j)$ is piecewise cubic on $[x_{n-1}, x_{n+1}]$, by splitting the interval in half, integration by Simpson's rule is exact. To evaluate the coefficients of $\hat{n}(j)$ and $\hat{n}(j) + 1$, the integral is similarly divided to account for the early exercise point, and a three point Gaussian quadrature is applied to obtain the coefficients in (52). $\square$

Remark 6 The coefficient recursion for the Bermudan put is initialized by the set terminal coefficients $\theta_k^M = \theta^M$ satisfying

$$\theta_n^M = \begin{cases} K - S_0 e^{x_n} \cdot (\vartheta_{[-1,0]} + \vartheta_{[0,1]}) & n = 1, \dots, \bar{n} - 1 \\ K \cdot (\frac{1}{2} - \vartheta_{[-1,0]}) & n = \bar{n} \end{cases} \quad (56)$$

where $\bar{n}$ is defined in equation (43), and $\theta_{k,n}^M = 0$ for $n = \bar{n} + 1, \dots, N/2$. This initial vector is the same for all $k = 1, \dots, m_0$.

3.1 Stochastic volatility models

In addition to Heston model given in equation (18), we consider the following group of stochastic volatility models, for which we provide the dynamics. Augmenting the model of Hull and White (1990) with jumps, we obtain

$$\text{HW: } \begin{cases} dS_t = S_t(r - \lambda\kappa)dt + S_t\sqrt{v_t}dW_t^1 + S_t(e^{J_t} - 1)dN_t, \\ dv_t = a_v v_t dt + \sigma_v v_t dW_t^2. \end{cases} \quad (57)$$

Extending the 4/2 model introduced in Grasselli (2016) to include jumps, we have

$$\text{4/2 Model: } \begin{cases} dS_t = S_t(r - \lambda\kappa)dt + S_t[a\sqrt{v_t} + \frac{b}{\sqrt{v_t}}]dW_t^1 + S_t(e^{J_t} - 1)dN_t, \\ dv_t = \eta(\theta - v_t)dt + \sigma_v \sqrt{v_t}dW_t^2. \end{cases} \quad (58)$$

The model of Stein and Stein (1991) is extended to include jumps and to allow correlation $\rho \in (-1, 1)$ between W_t^1 and W_t^2 :

$$\text{SS: } \begin{cases} dS_t = S_t(r - \lambda\kappa)dt + S_t v_t dW_t^1 + S_t(e^{J_t} - 1)dN_t, \\ dv_t = \eta(\theta - v_t)dt + \sigma_v dW_t^2. \end{cases} \quad (59)$$

We also consider the 3/2 model with jumps, which can be formulated as

$$\begin{cases} dS_t = S_t(r - \lambda\kappa)dt + S_t\sqrt{v_t}dW_t^1 + S_t(e^{J_t} - 1)dN_t, \\ dv_t = v_t[\eta(\theta - v_t)dt + \sigma_v \sqrt{v_t}dW_t^2]. \end{cases} \quad (60)$$

where $\sigma_v > 0$, $\theta \in \mathbb{R}$ is the mean reversion level, η is given such that ηv_t -a stochastic volatility quantity- is the speed of mean reversion. However, we prefer the alternative form which results from a change of variables

$$\widehat{v}_t = \frac{1}{v_t}, \quad \widehat{\eta} = \eta\theta, \quad \widehat{\theta} = \frac{\eta + \sigma_v^2}{\eta\theta}, \quad \widehat{\sigma}_v = -\sigma_v. \quad (61)$$

In particular, we use the alternative formulation

$$\mathbf{3/2 \text{ Model:}} \quad \begin{cases} dS_t = S_t(r - \lambda\kappa)dt + \frac{S_t}{\sqrt{\widehat{v}_t}}dW_t^1 + S_t(e^{J_t} - 1)dN_t, \\ d\widehat{v}_t = \widehat{\eta}[\widehat{\theta} - \widehat{v}_t]dt + \widehat{\sigma}_v\sqrt{\widehat{v}_t}dW_t^2, \quad \widehat{v}_0 = 1/v_0. \end{cases} \quad (62)$$

The model of Scott (1987) is extended to include jumps and to allow correlation $\rho \in (-1, 1)$ between W_t^1 and W_t^2 :

$$\mathbf{Scott:} \quad \begin{cases} dS_t = S_t(r - \lambda\kappa)dt + S_te^{v_t}dW_t^1 + S_t(e^{J_t} - 1)dN_t, \\ dv_t = \eta(\theta - v_t)dt + \sigma_v dW_t^2. \end{cases} \quad (63)$$

Given this formulation, v_t is interpreted as a stochastic driver, which determines instantaneous volatility through e^{v_t} . Unlike previous models, v_t is permitted to become negative.¹¹ A similar observation holds for the α -Hypergeometric model discussed next.

The α -Hypergeometric model (without jumps) was recently proposed by Da Fonseca and Martini (2016). Unlike Heston model, for α -Hypergeometric model the strict positivity of volatility is guaranteed. We extend their model by adding a jump component which results in the dynamics

$$\alpha\text{-H:} \quad \begin{cases} dS_t = S_t(r - \lambda\kappa)dt + S_te^{v_t}dW_t^1 + S_t(e^{J_t} - 1)dN_t, \\ dv_t = (\eta - \theta e^{av_t})dt + \sigma_v dW_t^2, \end{cases} \quad (64)$$

where $\eta, v_0 \in (-\infty, +\infty)$, $\theta > 0$, $\sigma_v > 0$, $a_v > 0$. For Scott model and the α -Hypergeometric model, we allow the underlying driver v_t to take negative values in the “variance” grid (see Remark 2).

Remark 7 While the mean and variance of v_t are unknown in the α -Hypergeometric model, we can estimate them using a simple approximation.¹² From the first order expansion

$$e^{av_t} \approx e^{av_0} + ae^{av_0}(v_t - v_0) = e^{av_0}(1 - av_0) + ae^{av_0}v_t$$

the dynamics of the variance process in the α -Hypergeometric model are approximated by

$$dv_t \approx \theta ae^{av_0} \left(\frac{\eta - \theta e^{av_0}(1 - av_0)}{\theta ae^{av_0}} - v_t \right) dt + \sigma_v dW_t^2 := \bar{\eta}(\bar{\theta} - v_t)dt + \sigma_v dW_t^2,$$

so we can estimate the mean and variance using Scott model as in Remark 2 and 3 with $t := T/2$.

¹¹The alternative is to consider an instantaneous volatility of the form τe^{v_t} for $\tau \in (0, 1)$, and restrict v_t .

¹²The accuracy of this approach has been verified with Monte Carlo simulation.

3.2 Error Analysis

We now show that the PROJ algorithm (with the linear basis) converges at a rate of $\mathcal{O}(\Delta^2)$ in the pure regime switching case. For concreteness, the result is demonstrated for a double barrier knock-out option, but extends to general barrier and Bermudan contracts, as discussed in Remark 8 below. Consider the case of a double barrier knock-out option (with knock-out region $\mathcal{C}^c$), where $\mathcal{C} = [l_x, u_x]$ with $u_x - l_x = \hat{\alpha}$, and recall $\mathcal{C}_n = \{\nu \in \mathbb{R} : x_n + \nu \in \mathcal{C}\}$ for grid point x_n . Denote the valuation error by

$$\mathcal{E}_m(x_n, j) := \mathcal{V}_m^*(x_n, j) - \mathcal{V}_m(x_n, j),$$

where $\mathcal{V}_m^*(x_n, j)$ is the true value, and $\mathcal{V}_m(x_n, j)$ is the approximation in the absence of projection coefficient error.¹³ In particular

$$\mathcal{V}_m(x_n, j) = e^{-r\Delta t} \sum_{k=1, \dots, m_0} P_{j,k}^{\Delta t} \int_{\mathcal{C}_n} \tilde{\mathcal{V}}_{m+1}(x_n + \nu, k) \hat{p}_{j,k}(\nu) d\nu$$

where $\mathcal{C}_n = \{\nu \in \mathbb{R} : x_n + \nu \in \mathcal{C}\}$, and

$$\hat{p}_{j,k}(\nu) := \sum_{n=1}^N \left(\int_{-\infty}^{\infty} p_{j,k}(\eta) \tilde{\varphi}_{a,n}(\eta) d\eta \right) \varphi_{a,n}(\nu) := \sum_{n=1}^N \beta_{a,n}^{j,k} \cdot \varphi_{a,n}(\nu).$$

Here $\tilde{\mathcal{V}}_{m+1}(x_n + \nu, k)$ is taken to be the quadratic interpolate value function.¹⁴ Hence

$$\begin{aligned} |\mathcal{E}_m(x_n, j)| &= e^{-r\Delta t} \left| \sum_{k=1, \dots, m_0} P_{j,k}^{\Delta t} \int_{\mathcal{C}_n} \left(\mathcal{V}_{m+1}^*(x_n + \nu, k) p_{j,k}(\nu) - \tilde{\mathcal{V}}_{m+1}(x_n + \nu, k) \hat{p}_{j,k}(\nu) \right) d\nu \right| \\ &\leq e^{-r\Delta t} \sum_{k=1, \dots, m_0} P_{j,k}^{\Delta t} \left| \int_{\mathcal{C}_n} \mathcal{V}_{m+1}^*(x_n + \nu, k) p_{j,k}(\nu) - \tilde{\mathcal{V}}_{m+1}(x_n + \nu, k) \hat{p}_{j,k}(\nu) d\nu \right| \\ &\leq e^{-r\Delta t} \sum_{k=1, \dots, m_0} P_{j,k}^{\Delta t} \cdot (|\mathcal{E}_{m+1}^1(x_n, j, k)| + |\mathcal{E}_{m+1}^2(x_n, j, k)|), \end{aligned} \quad (65)$$

where

$$\mathcal{E}_{m+1}^1(x_n, j, k) := \int_{\mathcal{C}_n} \mathcal{V}_{m+1}^*(x_n + \nu, k) p_{j,k}(\nu) - \mathcal{V}_{m+1}^*(x_n + \nu, k) \hat{p}_{j,k}(\nu) d\nu$$

and

$$\mathcal{E}_{m+1}^2(x_n, j, k) := \int_{\mathcal{C}_n} \mathcal{V}_{m+1}^*(x_n + \nu, k) \hat{p}_{j,k}(\nu) - \tilde{\mathcal{V}}_{m+1}(x_n + \nu, k) \hat{p}_{j,k}(\nu) d\nu.$$

¹³For simplicity, we omit the error in calculating the projection coefficients, as they converge at a much faster rate than the other sources, and does not affect the overall convergence rate (see Kirkby (2017b), which considers the influence of this error).

¹⁴To be more precise, the coefficients $\theta_{n,k}$ are determined by the quadratic interpolation $\tilde{\mathcal{V}}_{m+1}(x_n + \nu, k)$ over overlapping intervals $[x_{n-1}, x_{n+1}]$. However, the error bound is unaffected if we treat $\tilde{\mathcal{V}}_{m+1}(x_n + \nu, k)$ as a global quadratic interpolation (rather than a set of overlapping local interpolations), which simplifies the analysis. In fact, in the computation of $\theta_{n,k}$ we could alternatively use a global interpolation, but the local approach leads to closed-form coefficient formulas.

From the Cauchy-Schwartz inequality,

$$\begin{aligned} |\mathcal{E}_{m+1}^1(x_n, j, k)| &\leq \int_{\mathcal{C}_n} |\mathcal{V}_{m+1}^*(x_n + \nu, k)| |p_{j,k}(\nu) - \hat{p}_{j,k}(\nu)| d\nu \\ &\leq B_{\mathcal{V},1} \|p_{j,k} - \hat{p}_{j,k}\|_2, \end{aligned} \quad (66)$$

where $\int_{\mathcal{C}_n} |\mathcal{V}_{m+1}^*(x_n + \nu, k)|^2 d\nu = \int_{\mathcal{C}} |\mathcal{V}_{m+1}^*(y, k)|^2 d\nu \leq B_{\mathcal{V},1}^2$ independent of m for a double barrier contract (in general, we can take the finite bound over the grid $[x_1, x_n]$). Similarly,

$$\begin{aligned} |\mathcal{E}_{m+1}^2(x_n, j, k)| &\leq \int_{\mathcal{C}_n} p_{j,k}(\nu) |\mathcal{V}_{m+1}^*(x_n + \nu, k) - \tilde{\mathcal{V}}_{m+1}(x_n + \nu, k)| d\nu \\ &\quad + \int_{\mathcal{C}_n} |\hat{p}_{j,k}(\nu) - p_{j,k}(\nu)| |\mathcal{V}_{m+1}^*(x_n + \nu, k) - \tilde{\mathcal{V}}_{m+1}(x_n + \nu, k)| d\nu \\ &\leq \left\| \mathcal{V}_{m+1}^*(\cdot, k) - \tilde{\mathcal{V}}_{m+1}(\cdot, k) \right\|_{\infty}^{\mathcal{C}} \left(\int_{\mathcal{C}_n} p_{j,k}(\nu) d\nu + \int_{\mathcal{C}_n} |\hat{p}_{j,k}(\nu) - p_{j,k}(\nu)| d\nu \right) \\ &\leq \left\| \mathcal{V}_{m+1}^*(\cdot, k) - \tilde{\mathcal{V}}_{m+1}(\cdot, k) \right\|_{\infty}^{\mathcal{C}} \left(1 + \sqrt{\hat{\alpha}} \|p_{j,k} - \hat{p}_{j,k}\|_2 \right), \end{aligned} \quad (67)$$

where $\|g\|_{\infty}^{\mathcal{C}} := \sup\{g(x) : x \in \mathcal{C}\}$. Let $\mathcal{V}_{m+1,k}^*(x) := \mathcal{V}_{m+1}^*(x, k)$ and $\tilde{\mathcal{V}}_{m+1,k}(x) := \tilde{\mathcal{V}}_{m+1}(x, k)$. It follows that

$$\left\| \mathcal{V}_{m+1,k}^* - \tilde{\mathcal{V}}_{m+1,k} \right\|_{\infty}^{\mathcal{C}} \leq \left\| \mathcal{V}_{m+1,k}^* - \tilde{\mathcal{V}}_{m+1,k}^* \right\|_{\infty}^{\mathcal{C}} + \left\| \tilde{\mathcal{V}}_{m+1,k}^* - \tilde{\mathcal{V}}_{m+1,k} \right\|_{\infty}^{\mathcal{C}}$$

where $\tilde{\mathcal{V}}_{m+1,k}^*$ is the quadratic interpolation taken from the true values of $\tilde{\mathcal{V}}_{m+1,k}^*$. The first term satisfies

$$\begin{aligned} \left\| \mathcal{V}_{m+1,k}^* - \tilde{\mathcal{V}}_{m+1,k}^* \right\|_{\infty}^{\mathcal{C}} &\leq \frac{\Delta^3}{12} \cdot \left\| \mathcal{V}_{m+1,k}^{*(3)} \right\|_{\infty}^{\mathcal{C}} \\ &\leq \Delta^3 \cdot \frac{e^{-r\Delta t}}{12} \max_{j,k} \left\{ \left\| p_{k,j}^{(3)} \right\|_1^{(-\hat{\alpha}, \hat{\alpha})} \right\} \max_j \left\{ \left\| \mathcal{V}_{m+2,j}^* \right\|_{\infty}^{\mathcal{C}} \right\}, \end{aligned} \quad (68)$$

where the second inequality follows from

$$\begin{aligned} \left| \frac{d^n}{dx^n} \mathcal{V}_{m+1,k}^*(x) \right| &= \left| e^{-r\Delta t} \sum_{j=1, \dots, m_0} P_{k,j}^{\Delta t} \int_{\mathcal{C}} \mathcal{V}_{m+2,j}^*(y) \cdot p_{k,j}^{(n)}(y-x) dy \right| \\ &\leq e^{-r\Delta t} \sum_{j=1, \dots, m_0} P_{k,j}^{\Delta t} \left\| p_{k,j}^{(n)} \right\|_1^{(-\hat{\alpha}, \hat{\alpha})} \left\| \mathcal{V}_{m+2,j}^* \right\|_{\infty}^{\mathcal{C}} \\ &\leq e^{-r\Delta t} \max_{j,k} \left\{ \left\| p_{k,j}^{(n)} \right\|_1^{(-\hat{\alpha}, \hat{\alpha})} \left\| \mathcal{V}_{m+2,j}^* \right\|_{\infty}^{\mathcal{C}} \right\}, \end{aligned}$$

where $\sum_{j=1, \dots, m_0} P_{k,j}^{\Delta t} = 1$. From (68), we obtain

$$\max_k \left\| \mathcal{V}_{m+1,k}^* - \tilde{\mathcal{V}}_{m+1,k}^* \right\|_{\infty}^{\mathcal{C}} \leq \zeta_1 \Delta^3, \quad (69)$$

where

$$\zeta_1 := \frac{e^{-r\Delta t}}{12} B_{\mathcal{V},2} \cdot \max_{j,k} \left\{ \left\| p_{j,k}^{(3)} \right\|_1^{(-\hat{\alpha}, \hat{\alpha})} \right\}, \quad \zeta_2(\Delta) := \max_{j,k} \|p_{j,k} - \hat{p}_{j,k}\|_2$$

and $B_{\mathcal{V},2}$ satisfies $\max_j \left\{ \|\mathcal{V}_{m,j}^*\|_\infty^{\mathcal{C}} \right\} \leq B_{\mathcal{V},2}$ for all $0 \leq m \leq M$ since the true value function is finite on $\mathcal{C}$ (bounded by the maximum discounted payoff). For the second term, we have

$$\left\| \tilde{\mathcal{V}}_{m+1,k}^* - \tilde{\mathcal{V}}_{m+1,k} \right\|_\infty^{\mathcal{C}} \leq \max_j \left\{ \max_n |\mathcal{V}_{m+1,j}^*(x_n) - \mathcal{V}_{m+1,j}(x_n)| \right\} := \mathcal{E}_{m+1}^*. \quad (70)$$

Hence, with $\mathcal{E}_{m+1}^2 := \max_{n,j,k} |\mathcal{E}_{m+1}^2(x_n, j, k)|$, we have from (67), (69) and (70) that

$$\mathcal{E}_{m+1}^2 \leq (1 + \sqrt{\hat{\alpha}} \cdot \zeta_2(\Delta)) \cdot (\mathcal{E}_{m+1}^* + \zeta_1 \Delta^3). \quad (71)$$

Define $\mathcal{E}_{m+1}^1 := \max_{n,j,k} |\mathcal{E}_{m+1}^1(x_n, j, k)|$, which from (66) satisfies

$$\mathcal{E}_{m+1}^1 \leq B_{\mathcal{V},1} \cdot \zeta_2(\Delta). \quad (72)$$

Combining (65) with (71) and (72), we have for $0 \leq m \leq M-2$:

$$\begin{aligned} \mathcal{E}_m^* &\leq e^{-r\Delta_t} \sum_{k=1,\dots,m_0} P_{j,k}^{\Delta_t} \cdot (\mathcal{E}_{m+1}^1 + \mathcal{E}_{m+1}^2) \\ &\leq e^{-r\Delta_t} (\mathcal{E}_{m+1}^1 + \mathcal{E}_{m+1}^2) \\ &\leq e^{-r\Delta_t} (1 + \sqrt{\hat{\alpha}} \cdot \zeta_2(\Delta)) \cdot (\mathcal{E}_{m+1}^* + \zeta_1 \Delta^3) + e^{-r\Delta_t} B_{\mathcal{V},1} \cdot \zeta_2(\Delta) \\ &= \zeta_3(\Delta) \mathcal{E}_{m+1}^* + \zeta_4(\Delta), \end{aligned} \quad (73)$$

where

$$\zeta_3(\Delta) := e^{-r\Delta_t} (1 + \sqrt{\hat{\alpha}} \cdot \zeta_2(\Delta)), \quad \zeta_4(\Delta) := \zeta_3(\Delta) \cdot \zeta_1 \Delta^3 + e^{-r\Delta_t} B_{\mathcal{V},1} \cdot \zeta_2(\Delta).$$

At the first recursive pricing date, $M-1$, we have

$$|\mathcal{E}_{M-1}(x_n, j)| \leq e^{-r\Delta_t} \sum_{k=1,\dots,m_0} P_{j,k}^{\Delta_t} |\mathcal{E}_M^1(x_n, j, k)|, \quad (74)$$

since the terminal payoff is known, from which $\mathcal{E}_{M-1}^* \leq e^{-r\Delta_t} B_{\mathcal{V},1} \cdot \zeta_2(\Delta)$. Iterating backwards using (73), we obtain

$$\begin{aligned} \mathcal{E}_0^* &\leq (\zeta_3(\Delta))^{M-1} \mathcal{E}_{M-1}^* + \zeta_4(\Delta) \cdot \sum_{m=0}^{M-2} (\zeta_3(\Delta))^m \\ &\leq e^{-r\Delta_t} B_{\mathcal{V},1} \cdot \zeta_2(\Delta) \cdot (\zeta_3(\Delta))^{M-1} + \zeta_4(\Delta) \cdot \sum_{m=0}^{M-2} (\zeta_3(\Delta))^m. \end{aligned} \quad (75)$$

The dominant term in $\zeta_4(\Delta)$ is $\zeta_2(\Delta) = \max_{j,k} \|p_{j,k} - \hat{p}_{j,k}\|_2 = \mathcal{O}(\Delta^2)$ from Kirkby (2015) (the theoretical convergence rate of frame projection), and for Δ sufficiently small, $\zeta_3(\Delta) \approx 1$, from which the second term in (75) decays like $\mathcal{O}(\Delta^2)$. The same holds for the first term, and we conclude that the overall rate of convergence is $\mathcal{O}(\Delta^2)$.

Remark 8 In practice, option price approximations using frame projection converge at a much faster rate than predicted, as illustrated in Table 3. By increasing the number of basis elements N , the projection error can be controlled. For Bermudan options, there is also the error in locating $\hat{x}^m(j)$, which is again $\mathcal{O}(\Delta^2)$ using linear interpolation,¹⁵ and does not affect the overall rate of convergence. While the proof above was presented for a double barrier option, the more general case of unbounded continuation region $\mathcal{C}$ can be proved using the grid extension developed in Kirkby (2017b). The method uses an approximation of the value function at the boundaries, and allows for a smaller value of $\hat{\alpha}$ when truncating $\mathcal{C}$ (recall Remark 5). We refer the reader to the previous reference for details, which can be readily adapted to the present case.

Remark 9 For stochastic volatility models, there is the additional error induced by a Markov chain approximation of the underlying variance process. We find empirically that an algebraic order of convergence holds with respect to m_0 , the number of variance states. Numerical experiments demonstrating the convergence across a wide set of models are provided in Sections 4.3 and 5, where we use benchmark prices from Monte Carlo as well as three additional numerical methods.

4 Numerical Examples

4.1 Some jump distributions

Recall from (2) that the quantity $e^{\gamma(y,i)}$ represents the jump amplitude associated with $\nu_i(y)$. In the remainder of the paper, as a specific example, we take $\gamma(y,i) = y$ for all $i \in \mathcal{M}$. We list below several popular models for the distribution of the random jump size Y .

Normal distribution: In this case, $\nu_j(y)$ is the density of the $\text{Normal}(a_j, b_j^2)$ distribution, i.e. $e^{\gamma(Y,j)}$ is log-normal. This jump model is first considered in Merton (1976). Under this model, $\kappa_j = \exp(a_j + \frac{1}{2}b_j^2) - 1$. We then have $\phi_j(z) = e^{iza_j - \frac{1}{2}z^2b_j^2}$ and with the jump rate λ_j for $j = 1, 2, \dots, m_0$,

$$\psi_j(z) = iz\zeta_j - \frac{1}{2}z^2\sigma_j^2 + \lambda_j \left(e^{iza_j - \frac{1}{2}z^2b_j^2} - 1 \right), \quad j = 1, 2, \dots, m_0. \quad (76)$$

However, this distribution has several major drawbacks: (i) the jumps cannot be large, hence most of the jumps are not easily detected, and (ii) the return process $\log(S_{t+h}/S_t)$ lacks the observed leptokurtic properties, i.e. fat tails, in the empirical data.

Double-exponential distribution: This model is proposed in Kou (2002). In this case,

$$\nu_j(y) = p_j\eta_{1,j}e^{-\eta_{1,j}y}\mathbb{I}_{\{y \geq 0\}} + (1 - p_j)\eta_{2,j}e^{\eta_{2,j}y}\mathbb{I}_{\{y < 0\}}, \quad i = 1, \dots, m_0. \quad (77)$$

¹⁵Higher order schemes may be used, but are not found to improve convergence.

Here $p_j \in (0, 1)$, $\eta_{1,j} > 1$, $\eta_{2,j} > 0$ are assumed to ensure the finiteness of $\mathbb{E}(e^Y)$ and $\mathbb{E}(S_t)$. Then we have $\phi_j(z) = p_j \frac{\eta_{1,j}}{\eta_{1,j} - iz} + (1 - p_j) \frac{\eta_{2,j}}{\eta_{2,j} + iz}$ and with the jump rate λ_j for $j = 1, 2, \dots, m_0$,

$$\psi_j(z) = iz\zeta_j - \frac{1}{2}z^2\sigma_j^2 + \lambda_j \left(p_j \frac{\eta_{1,j}}{\eta_{1,j} - iz} + (1 - p_j) \frac{\eta_{2,j}}{\eta_{2,j} + iz} - 1 \right), \quad j = 1, 2, \dots, m_0. \quad (78)$$

This model can better explain the asymmetric leptokurtic feature and volatility smile observed from empirical data. Under this model $\kappa_i = p_i \frac{\eta_{1,i}}{\eta_{1,i} - 1} + (1 - p_i) \frac{\eta_{2,i}}{\eta_{2,i} + 1} - 1$.

Mixture of two normal distributions: In Florescu et al. (2013), a jump distribution as a mixture of two normal distribution is proposed. In this case, $\nu_i(y)$ has the following form:

$$\nu_j(y) = p_j \frac{1}{\sqrt{2\pi}b_{1,j}} e^{-\frac{(y-a_{1,j})^2}{2b_{1,j}^2}} + (1 - p_j) \frac{1}{\sqrt{2\pi}b_{2,j}} e^{-\frac{(y-a_{2,j})^2}{2b_{2,j}^2}}, \quad i = 1, \dots, m_0, \quad (79)$$

where $a_{1,j} > 0$, $a_{2,j} < 0$, and $0 < p_j < 1$. Under this model, $\kappa_j = p_j e^{a_{1,j} + \frac{1}{2}b_{1,j}^2} + (1 - p_j) e^{a_{2,j} + \frac{1}{2}b_{2,j}^2} - 1$, where $p_j \in (0, 1)$ and $(1 - p_j)$ represent the upward and downward jumps, respectively. Then we have

$$\phi_j(z) = p_j e^{iza_{1,j} - \frac{1}{2}z^2b_{1,j}^2} + (1 - p_j) e^{iza_{2,j} - \frac{1}{2}z^2b_{2,j}^2},$$

and with the jump rate λ_j for $j = 1, 2, \dots, m_0$,

$$\psi_j(z) = iz\zeta_j - \frac{1}{2}z^2\sigma_j^2 + \lambda_j (\phi_j(z) - 1), \quad j = 1, 2, \dots, m_0. \quad (80)$$

This model can be considered as a distribution having probability p_j to jump up with the jump size drawn from a normal distribution with parameters $a_{1,j}$ and $b_{1,j}^2$, and with probability $(1 - p_j)$ to jump down with the jump size drawn from a normal distribution with parameters $a_{2,j}$ and $b_{2,j}^2$ in regime j .

4.2 American put options in a regime switching model

In this section, we consider Bermudan and American option pricing within regime switching jump diffusion models. For pricing American options with PROJ, we use a two point Richardson extrapolated price: $2 \cdot P(2 \cdot M_1) - P(M_1)$ where $P(M_1)$ is the Bermudan price with M_1 monitoring dates. Bermudan options are priced with a fixed number of monitoring dates M .

To assess the convergence rate of PROJ for regime switching models, we use the regime switching PROJ algorithm to price (single state) Lévy models for which reference prices are available in the literature. In this case, we set each regime to be consistent with the single state of the Lévy process. The errors are reported in Table 3 as a function of monitoring dates M , and log return grid size N , for each of three test sets. The first set of tests are for the BSM model:

Test 1 (BSM): $S_0 = 100$, $K = 110$, $r = 0.1$, $q = 0.0$, $T = 1$, $\sigma = 0.20$

with reference price for $M = 10$ obtained by the COS method of Fang and Oosterlee (2009), and for $M = 50, 250$ prices are obtained by the PROJ method for Lévy processes of Kirkby (2017a). Errors are reported for PROJ by pricing the two state RS model with Q defined by $q_{12} = 6, q_{21} = 9$, as in Khaliq and Liu (2009). The second set of tests are for the BSM model:

$$\text{Test 2 (BSM): } S_0 = 100, K = 110, r = 0.1, q = 0.0, T = 1, \sigma = 0.25$$

with reference price for $M = 10$ obtained by the CONV method of Lord et al. (2008), and for $M = 50, 250$ prices are obtained by the PROJ method. For Test 2 and Test 3 we use the three state RS model with Q defined in equation (81), as considered by Rambeericha and Pantelous (2016). The third set of tests are for the Normal Inverse Gaussian model (NIG):

$$\text{Test 3 (NIG): } S_0 = 100, K = 100, r = 0.05, q = 0.02, T = 1, \alpha = 15, \beta = -5, \delta = 0.5$$

which has a risk neutral Lévy symbol (in each state) defined by

$$\psi(\xi) = -\delta \left(\sqrt{\alpha^2 - (\beta + i\xi)^2} - \sqrt{\alpha^2 - \beta^2} \right) + i\xi \left(r - q + \delta \left(\sqrt{\alpha^2 - (\beta + 1)^2} - \sqrt{\alpha^2 - \beta^2} \right) \right).$$

Reference prices for $M = 12, 52, 252$ are obtained by the Hilbert transform (HT) method of Feng and Lin (2013). From Table 3 we see that the regime switching PROJ algorithm achieves very high accuracy for pricing Bermudan options, and convergence is easily obtained. For the NIG model with large M , a larger value of N is required than for the BSM cases, due to its highly peaked density for small Δ_t . In addition to N , the only remaining parameter to apply the PROJ method is $\hat{\alpha}$ (which determines log asset grid width), for which we set $\hat{\alpha} = 4$ for $M \geq 50$ and $\hat{\alpha} = 6$ for $M < 50$. When obtaining references prices, a larger value of $\hat{\alpha}$ may be set.

$\log_2(N)$	9	10	11	12	13	Ref.	Source	M
Test 1	1.70e-05	9.46e-07	8.47e-08	3.31e-09	1.77e-11	10.479520123	COS	10
	7.37e-05	1.54e-07	3.36e-07	1.89e-08	4.12e-10	10.678044064	PROJ	50
	5.99e-05	1.30e-05	1.65e-06	4.08e-08	3.32e-09	10.710838317	PROJ	250
Test 2	7.69e-06	4.54e-07	1.66e-08	1.58e-09	1.42e-10	11.98745352	CONV	10
	9.64e-07	4.97e-07	7.74e-08	3.03e-09	9.90e-12	12.132564348	PROJ	50
	8.33e-06	2.47e-06	2.56e-07	1.90e-08	1.00e-09	12.162005425	PROJ	250
Test 3	2.30e-05	5.73e-07	3.36e-07	1.48e-08	7.28e-10	6.4574297377	HT	12
	4.34e-04	7.94e-06	1.63e-07	2.85e-09	3.88e-09	6.4833874148	HT	52
	6.49e+00	1.24e-02	3.15e-04	5.00e-06	8.87e-09	6.489580997740	HT	252

Table 3: Regime Switching Convergence. Bermudan options pricing in standard (non-regime switching) Lévy models using the PROJ regime switching algorithm. Reference price sources are listed in the “Source” column.

In Table 4, we consider a two-state regime switching diffusion model with no jumps. We choose the parameters as: $Q = \begin{pmatrix} -6 & 6 \\ 9 & -9 \end{pmatrix}$, $K = 9$, $r = 0.05$, $\sigma_1 = 0.80$ $\sigma_2 = 0.30$. For comparison, we report the results using the finite difference scheme of Khaliq and Liu (2009) (in “PDE” columns) and the tree method of Jiang et al. (2016) (in “Tree” columns). The results obtained by our method are reported in PROJ columns. In particular, all three methods produce very similar results, and the main source of error in this case is the observed slow convergence of the Bermudan approximations to the American price, given the high volatility in each state.

S_0	V_1			V_2		
	PROJ	PDE	Tree	PROJ	PDE	Tree
6.0	3.5286	3.5358	3.5325	3.4505	3.4583	3.4534
7.5	2.7084	2.7143	2.7124	2.6133	2.6196	2.6159
8.5	2.2773	2.2825	2.2815	2.1759	2.1812	2.1784
9.0	2.0906	2.0954	2.0950	1.9873	1.922	1.9900
9.5	1.9208	1.9252	1.9253	1.8165	1.820	1.8193
10.5	1.6257	1.6295	1.6306	1.5214	1.5253	1.5245
12.0	1.2745	1.2776	1.2803	1.1742	1.1773	1.1782

Table 4: American puts in a 2-state regime switching diffusion model. Parameters: $K = 9$, $r = 0.05$, $\sigma_1 = 0.80$, $\sigma_2 = 0.30$, $T = 1$, $q_{12} = 6$, $q_{21} = 9$. V_j denotes the value of the put option in the regime j . PROJ: $N = 2^{12}$, $\hat{\alpha} = 6$, $M_1 = 300$ with Richardson extrapolation.

Next, we consider American put options in a 2-state regime switching jump diffusion model, with jumps modeled by a mixture of normal and by a double exponential distribution. We choose parameters as in Jiang et al. (2016): the generator $Q = \begin{pmatrix} -0.5 & 0.5 \\ 0.5 & -0.5 \end{pmatrix}$, $K = 100$, $r = 0.05$, $\sigma_1 = 0.15$ $\sigma_2 = 0.25$. For comparison, we report in Table 5 the results obtained by the tree method of Jiang et al. (2016) (in Tree columns) using 100 time step size.

S_0	Mixture of Normal Jumps				Double Exponential Jumps			
	Regime 1		Regime 2		Regime 1		Regime 2	
	PROJ	Tree	PROJ	Tree	PROJ	Tree	PROJ	Tree
92	47.4172	47.5846	36.7820	36.8430	36.0193	36.0820	28.0313	28.0308
96	46.4487	46.6169	35.6286	35.6893	34.5986	34.6621	26.4319	26.4363
100	45.5286	45.6973	34.5529	34.6133	33.2572	33.3213	24.9477	24.9504
104	44.6528	44.8220	33.5478	33.6084	31.9913	32.0561	23.5727	23.5747
108	43.8178	43.9874	32.6068	32.6679	30.7967	30.8622	22.3001	22.3021

Table 5: American puts in regime switching jump diffusion model. Parameters: $T = 1, K = 100, r = 0.05, \sigma_1 = 0.15, \sigma_2 = 0.25, p_1 = p_2 = 0.3445, \lambda_1 = 5, \lambda_2 = 2$. Mixture of two normal jumps: $a_{1,1} = a_{1,2} = 0.3753, a_{2,1} = a_{2,2} = -0.5503, b_{1,1} = b_{1,2} = 0.18, b_{2,1} = b_{2,2} = 0.6944$. DE jumps: $\eta_{1,1} = \eta_{1,2} = 3.0465, \eta_{2,1} = \eta_{2,2} = 3.0775$. PROJ: $N = 2^{12}, \hat{\alpha} = 8, M_1 = 300$ with Richardson Extrapolation

Next, we compare our method with the finite scheme of Rambeericha and Pantelous (2016), in a case of a three regime model with normally distributed jumps (i.e., in each state the process evolves following the Merton's jump diffusion). The generator of the Markov chain $\alpha(t)$ is given by

$$Q = \begin{pmatrix} -0.8 & 0.6 & 0.2 \\ 0.2 & -1.0 & 0.8 \\ 0.1 & 0.3 & -0.4 \end{pmatrix}, \quad (81)$$

and the arrival rate of jumps differs in each state.

In Table 6, we report the results obtained in Rambeericha and Pantelous (2016) with different sets of space step (M) and time step (N) in RP columns. The results obtained by our method are reported in the PROJ row, where Richardson extrapolation is used to obtain the American option price. In particular, the results agree in each regime to within $10^{-5} \sim 10^{-6}$.

	Regime 1			Regime 2			Regime 3		
	M	N	RP	M	N	RP	M	N	RP
	64	128	11.1261351	64	128	13.8328230	64	128	15.7532644
	128	256	11.1250693	128	256	13.8314890	128	256	15.7516936
	256	512	11.1250406	256	512	13.8313990	256	512	15.7515986
PROJ	-	-	11.1250534	-	-	13.8314058	-	-	15.7516115
Diff	-	-	1.2800e-05	-	-	6.8000e-06	-	-	1.2900e-05

Table 6: American put option under a 3-regime jump diffusion model. Parameters: $K = S_0 = 100, T = 1.0, r = 0.05, \sigma_1 = \sigma_2 = \sigma_3 = 0.15$. Normal jumps for 3 regimes with $a_i = -0.5, b_i = 0.45, i = 1, 3, \lambda_1 = 0.3, \lambda_2 = 0.5, \lambda_3 = 0.7$. PROJ with Richardson extrapolation: $M_1 = 128, N = 2^{12}$.

Table 7 reports the computational cost of our proposed algorithm in seconds, taken as an average over 100 trials for each parameter setting, where all experiments are conducted in Matlab 8.5 on a personal computer with Intel(R) Core(TM) i7-6700 CPU @3.40GHz. In particular,

we consider the cost as a function of the key parameters: m_0 , M , and N . For comparison, $m_0 = 1$ illustrates the cost of the (single state) Lévy pricing algorithm of Kirkby (2017a). The additional initialization cost of the matrix exponential $\tilde{\mathcal{E}}$ is clear for $m_0 = 2, 3$. With m_0 fixed, the cost grows linearly with respect to M , and at the rate $N \log_2(N)$ as we increase N . For stochastic volatility models, considered next, the main cost becomes initialization of the matrix exponentials, which grows at a rate of m_0^3 . However, accurate valuations are obtained with a relatively small value of m_0 .

m_0	$M = 12$			$M = 52$			$M = 252$		
	1	2	3	1	2	3	1	2	3
9	0.0004	0.0698	0.0768	0.0017	0.0679	0.0797	0.0056	0.0768	0.1080
$\log_2(N)$ 10	0.0005	0.1408	0.1555	0.0017	0.1384	0.1591	0.0076	0.1541	0.2038
11	0.0008	0.2894	0.3392	0.0027	0.2904	0.3480	0.0121	0.3200	0.4248

Table 7: CPU times in seconds for Bermudan option pricing in regime switching model averaged over 100 trials each, with total log return space grid budget N .

From Tables 3 to Table 6, it can be seen that the method we propose can be used to compute the value of Bermudan and American options within regime switching models with/without jumps with very high accuracy. In the following sections, we will consider pricing American put options and barrier options in stochastic volatility models. Moreover, the method requires just milliseconds, as illustrated in Table 7.

4.3 Heston model

In this section, we consider the problem of pricing options in Heston model using the proposed method. First, we consider the price of European call options for which reference prices are available. The results obtained by our method, the COS method of Fang and Oosterlee (2008), and the tree method of Liu and Nguyen (2015) are reported as PROJ, COS, and Tree columns in Table 8 respectively, where European option prices for PROJ are computed using the recursive barrier algorithm with an appropriate barrier. While COS and PROJ agree in all cases within 10^{-6} or better, the tree method is less accurate, especially for a large v_0 .

S_0	$v_0 = 0.04$			$v_0 = 0.09$		
	PROJ	COS	Tree	PROJ	COS	Tree
90	0.8852	0.8852	0.8852	1.9024	1.9024	1.9017
95	2.2588	2.2588	2.2587	3.6147	3.6147	3.6140
100	4.6105	4.6105	4.6106	6.0703	6.0703	6.0695
105	7.9290	7.9290	7.9290	9.2385	9.2385	9.2372
110	12.0006	12.0006	12.0007	13.0088	13.0088	13.0061

Table 8: European call options under Heston model (without jumps). Parameters: $K = 100, r = 0.05, \eta = 3, \theta = 0.04, \rho = -0.1, \sigma_v = 0.1, T = 0.25$. PROJ values obtained using DOC barrier algorithm with small $L = 40$: $m_0 = 80, \gamma = 3.3, N = 2^{10}, \hat{\alpha} = 4$.

In Figure 1, we consider two sets of experiments using the PROJ barrier algorithm to price European options under Heston model. In general, the same convergence behavior holds for European, Bermudan, and barrier options. Focusing here on the European case, we can obtain precise benchmark prices for comparison using the COS method of Fang and Oosterlee (2008) for European options. The prices are verified up to 14 decimals by the PROJ method for European options.

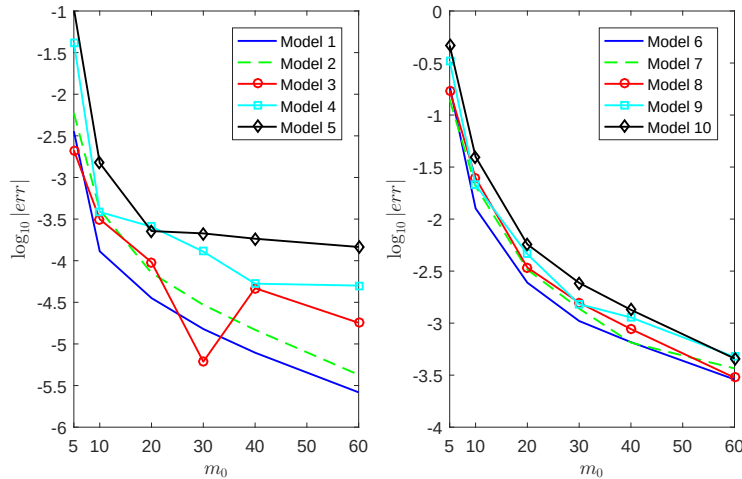


Figure 1: Challenging parameter sets: convergence of DOC barrier algorithm for European calls in Heston model as function of m_0 . Parameters: $S_0 = W = 50, r = 0.05, T = .25$. PROJ with $N = 2^{12}, M = 12, \hat{\alpha} = 6, \gamma = 3.3$.

For Models 1-5, $\eta = 3, \rho = -0.6$. We then set $v_0 = \theta$ for each, where respectively $v_0 = [0.01, 0.02, 0.03, 0.04, 0.05]$ and $\sigma_v = [0.02, 0.08, 0.14, 0.20, 0.26]$. For Models 6-10, we fix $\sigma_v = 0.15, \theta = 0.02, v_0 = 0.04$, and respectively $\eta = [1, 0.5, 0.25, 0.1, 0.05]$ and $\rho = [-0.9, -0.8, -0.7, -0.6, -0.5]$. The left panel in Figure 1 aims to capture the effect of increasing variance level on obtained accuracy. For Models 1-5, initial and long term variance (v_0, θ) are increasing, as well as the volatility of variance, σ_v . Overall, with m_0 fixed accuracy is greater for a smaller, more contained variance process, although practical accuracy of $10^{-3} \sim 10^{-4}$ is easily obtained in general.

The right panel of Figure 1 illustrates accuracy as a function of the quantity $q := 2\eta\theta/(\sigma_v^2) - 1$, which is found to be $[0.78, -0.11, -0.56, -0.82, -0.91]$ respectively for Models 6 – 10. It is well known that $q \in [-1, 0]$ results in a very difficult parameter set numerically, especially as q approaches -1 . However, the method easily obtains practical accuracy in each of these cases, which are chosen for their difficulty. A detailed investigation of near-singular behavior of variance as q approaches -1 can be found in Fang and Oosterlee (2011), for which they propose a novel solution. As illustrated in Figure 1, our method is also able to handle challenging parameter sets, despite its generality.

	S_0	8	9	10	11	12	CPU (sec)
	ref.val.	2.000000	1.107621	0.520030	0.213677	0.082044	-
$M = 10$	PROJ	0.00e+00	4.77e-03	3.20e-03	1.39e-03	4.37e-04	1.02
	COS	1.80e-02	4.79e-03	2.85e-03	1.31e-03	5.18e-04	6.9
	HT	1.80e-02	4.75e-03	2.81e-03	1.29e-03	5.14e-04	6.6
$M = 20$	PROJ	0.00e+00	2.43e-03	1.84e-03	7.84e-04	2.04e-04	1.20
	COS	9.54e-03	2.39e-03	1.40e-03	6.65e-04	2.78e-04	7.5
	HT	9.53e-03	2.31e-03	1.31e-03	6.07e-04	2.53e-04	7.2
$M = 40$	PROJ	0.00e+00	3.08e-03	2.84e-03	8.53e-04	1.19e-04	1.55
	COS	5.14e-03	1.07e-03	5.50e-04	2.54e-04	1.22e-04	8.9
	HT	5.12e-03	9.11e-04	3.60e-04	1.27e-04	5.40e-05	8.4
$M_1 = 8$	PROJ	0.00e+00	1.11e-04	5.03e-04	2.09e-04	8.40e-06	2.11

Table 9: American option value convergence from Bermudan estimates as a function of M . Richardson extrapolation estimate with $M_1 = 8$. Parameters: $K = 10, T = 0.25, r = 0.1, \eta = 5, \sigma_v = 0.9, \theta = 0.16, v_0 = 0.0625$. PROJ: $N = 2^{10}$, $\hat{\alpha} = 4$, $m_0 = 30$. Ref. values from COS.

In Table 9, we replicate an American option pricing experiment in Heston model from Fang and Oosterlee (2011) for which American reference values are provided.¹⁶ The rows $M = 10, 20, 40$ illustrate the price discrepancy for a Bermudan approximation using the PROJ algorithm, as compared with the COS method and the Hilbert Transform (HT) method of Zeng and Kwok (2014). The row $M_1 = 8$ illustrates the price error for the Richardson extrapolated value with $M_1 = 8$. We observe that practical accuracy is attained in each case and for all S_0 , at a negligible cost. It is also interesting to note that the pricing error for each S_0 is nearly identical to the errors reported by Fang and Oosterlee (2011) and (indicating that the dominant source of error here is likely time discretization), with the exception $S_0 = 8$, for which we match the reference value exactly. In this case, the optimal policy is to exercise immediately (ie at t_0), and the (intrinsic) value is just the payoff $K - S_0$.

¹⁶CPU times for the COS method are reported by the authors with an Intel(R) 2.2GHz CPU and 4-GB memory in MATLAB.

Next, we consider American put options under Heston model with jumps. We compare our results with those obtained by using IMEX scheme of Salmi et al. (2014) and the tree method in Table 10. For $S_0 = 90$, the prices obtained are similar for each, while for $S_0 = 100, 110$ the tree method diverges from the other two. This divergence was also observed for the tree method when pricing regime switching models in the presence of jumps.

S_0	PROJ	IMEX	Tree
90	11.6171	11.6199	11.6104
100	6.7295	6.7142	6.5267
110	4.2192	4.2616	4.3866

Table 10: American put options under Heston model. Parameters: $K = 100, r = 0.03, \eta = 2, \theta = 0.04, \rho = -0.50, \sigma_v = 0.25, v_0 = 0.04, T = 0.5$. Normal jumps: $a_1 = -0.5, b_1 = 0.04, \lambda = 0.20$. PROJ: $N = 2^{11}, m_0 = 40, \hat{\alpha} = 4, M_1 = 30$ with Richardson Extrapolation.

An additional set of reference prices is provided in Table 11 for Heston model without jumps, as well as for three jump distributions. For comparison, we also supply prices for the Tree method. In the absence of jumps, the prices are closely matched, often to four decimals. Prices also match closely in the case of jumps, but to a lesser extent.

K	No Jumps		Normal Jumps		DE Jumps		Mixture NJ	
	PROJ	Tree	PROJ	Tree	PROJ	Tree	PROJ	Tree
40	0.0160	0.0160	0.5851	0.5826	0.3529	0.3497	0.0338	0.0311
45	0.2872	0.2872	1.1998	1.1970	0.8367	0.8311	0.3605	0.3538
50	1.7376	1.7376	2.7374	2.7524	2.3039	2.3082	1.8434	1.8433
55	5.1605	5.1604	5.8099	5.8413	5.4481	5.4572	5.2095	5.2116
60	10.0000	10.0000	10.1026	10.1137	10.0000	10.0000	10.0000	10.0000

Table 11: American puts in Heston model. Parameters: $S_0 = 50, r = 0.05, T = 0.25, \rho = -0.10, v_0 = 0.04, \sigma_v = 0.10, \eta = 3, \theta = 0.04$. Normal Jumps $a_1 = -0.10, b_1 = 0.30, \lambda = 1$. Mixture of two normal jumps: $a_1 = -0.05, b_1 = 0.07, a_2 = 0.02, b_2 = 0.03, p = 0.6, \lambda = 1$. DE jumps: $\eta_1 = 10, \eta_2 = 5, p = 0.40, \lambda = 1$. PROJ: $N = 2^{11}, m_0 = 60, \hat{\alpha} = 6, M_1 = 30$ with Richardson Extrapolation.

5 Other stochastic volatility models

In this section we consider several additional stochastic volatility models and provide a large set of reference prices for Bermudan and barrier options. For many of these models (and their jump extensions), Bermudan/American and barrier option prices have not yet been reported in the existing literature. Table 12 provides two test parameter settings for each stochastic

volatility model, while Table 13 provides a set of jump distributions that will be considered for each set of parameters.

Model	Test Set 1						Test Set 2					
	σ_v	η	ρ	θ	v_0	a_v	σ_v	η	ρ	θ	v_0	a_v
SS	0.15	3	-0.6	0.18	0.15	-	0.18	2	-0.5	0.18	0.22	-
HW	0.15	-	-0.6	-	0.04	0.01	0.10	-	-0.7	-	0.03	0.03
3/2	0.15	4	-0.6	0.03	0.03	-	0.12	2	-0.3	0.04	0.035	-
4/2	0.15	0.5	-0.6	0.035	0.04	-	0.10	1.8	-0.7	0.04	0.04	-
Hes	0.15	4	-0.6	0.035	0.04	-	0.12	1.5	-0.8	0.035	0.04	-
Scott	0.15	3	-0.6	$\log(0.15)$	$\log(0.15)$	-	0.20	2	-0.9	$\log(0.16)$	$\log(0.18)$	-
α -H	0.25	0.01	-0.6	0.04	$\log(0.19)$	0.01	0.20	0.05	-0.9	0.2	$\log(0.17)$	0.03

Table 12: Parameters for stochastic volatility tests. In all cases, $r = 0.05$, $S_0 = 100$. For the 4/2 Model: $a = 0.5$, $b = 0.5 \cdot v_0$.

Jump Model	Parameters
Normal	$a_1 = -0.10, b_1 = 0.30, \lambda = 1$
Mixed Normals	$a_1 = -0.10, b_1 = 0.30, a_2 = 0.2, b_2 = 0.15, p = 0.6, \lambda = 1$
Double Exponential	$\eta_1 = 10, \eta_2 = 5, p = 0.40, \lambda = 1$

Table 13: Jump distribution parameters for numerical experiments

The first set of tests considers down-and-out barrier put options in stochastic volatility models without jumps. To provide a benchmark for models (given a lack of prices reported in the literature for most models), we calculate prices in Table 14 and 15 using a simple Euler Monte Carlo (MC) scheme with 5,000,000 sample paths, and time partitioning to reduce bias. The estimates from Monte Carlo are generally within a couple of cents of PROJ.¹⁷ In Table 16, we calculate prices in the two test sets with the addition of a double exponential jump component.

¹⁷The choice of Monte Carlo scheme here is simply to provide a benchmark, rather than as a performance comparison, in which case more sophisticated techniques could be employed.

K	Test Set 1 (PROJ)					Test Set 1 (MC)				
	90	95	100	105	110	90	95	100	105	110
SS	0.3065	0.9559	2.1872	4.1594	6.9423	0.3067	0.9569	2.1915	4.1715	6.9663
HW	0.4380	1.2896	2.7593	4.8973	7.6707	0.4332	1.2800	2.7466	4.8851	7.6620
3/2	0.4134	1.2798	2.8455	5.1852	8.2507	0.4125	1.2794	2.8475	5.1929	8.2654
4/2	0.4550	1.3388	2.8562	5.0459	7.8611	0.4569	1.3409	2.8588	5.0476	7.8608
Hes	0.3847	1.1607	2.5430	4.6104	7.3524	0.3840	1.1605	2.5443	4.6158	7.3651
Scott	0.3329	1.1052	2.6125	4.9931	8.2130	0.3303	1.1010	2.6088	4.9927	8.2188
α -H	0.3830	1.1528	2.5275	4.5923	7.3405	0.3816	1.1512	2.5283	4.5988	7.3546

Table 14: Down-and-out barrier puts under Stochastic Volatility (No Jumps). Parameters are Test Set 1 as in Table 12. $T = 0.5$, $M = 30$. PROJ: $N = 2^{13}$, $\hat{\alpha} = 6$, $m_0 = 120$. Knock out $L = 80$.

K	Test Set 2 (PROJ)					Test Set 2 (MC)				
	90	95	100	105	110	90	95	100	105	110
SS	0.3096	0.9312	2.0578	3.7923	6.1750	0.3107	0.9351	2.0668	3.8108	6.2064
HW	0.4128	1.2708	2.8170	5.1299	8.1698	0.4072	1.2574	2.7965	5.1044	8.1422
3/2	0.4409	1.3241	2.8719	5.1382	8.0774	0.4410	1.3259	2.8773	5.1505	8.0991
4/2	0.4525	1.3326	2.8456	5.0325	7.8496	0.4543	1.3359	2.8484	5.0330	7.8458
Hes	0.3688	1.1058	2.4195	4.3950	7.0373	0.3701	1.1092	2.4278	4.4116	7.0650
Scott	0.3705	1.1510	2.5839	4.7725	7.7061	0.3710	1.1538	2.5916	4.7873	7.7295
α -H	0.3473	1.0917	2.4888	4.6674	7.6334	0.3461	1.0918	2.4923	4.6767	7.6514

Table 15: Down-and-out barrier puts under Stochastic Volatility (No Jumps). Parameters are Test Set 2 as in Table 12. $T = 0.5$, $M = 30$. PROJ: $N = 2^{13}$, $\hat{\alpha} = 6$, $m_0 = 120$. Knock out $L = 80$.

K	Test Set 1					Test Set 2				
	90	95	100	105	110	90	95	100	105	110
SS	0.2576	0.7802	1.7411	3.2543	5.3920	0.2582	0.7637	1.6648	3.0410	4.9357
HW	0.3324	0.9787	2.1046	3.7684	5.9707	0.3101	0.9460	2.0969	3.8478	6.2076
3/2	0.3095	0.9478	2.1078	3.8744	6.2523	0.3289	0.9858	2.1493	3.8862	6.1944
4/2	0.3411	1.0052	2.1603	3.8597	6.0955	0.3394	1.0013	2.1535	3.8507	6.0867
Hes	0.3001	0.8971	1.9595	3.5625	5.7241	0.2926	0.8685	1.8903	3.4339	5.5247
Scott	0.2638	0.8383	1.9386	3.6949	6.1397	0.2897	0.8837	1.9661	3.6319	5.9081
α -H	0.3008	0.8952	1.9530	3.5527	5.7161	0.2794	0.8533	1.9110	3.5618	5.8474

Table 16: Down-and-out barrier puts under Stochastic Volatility with double exponential jumps. Parameters as in Table 12. $T = 0.5$, $M = 30$. PROJ: $N = 2^{13}$, $\hat{\alpha} = 6$, $m_0 = 120$. Knock out $L = 80$.

To assess the efficiency of our approach, we compute the CPU times averaged over 20 trials of each parameter setting in Table 17, as a function of the key parameters: m_0 , N , and M which drive the computational cost. These time estimates are provided for Bermudan options, while barrier options are somewhat less expensive to compute. Since the times are essentially identical for each different model, we use Heston model (with normal jumps) as an example. A value of $m_0 = 10 \sim 20$ is often sufficient to obtain a quick price estimate, valid to within several cents of the true price. Greater accuracy of about $10^{-3} \sim 10^{-5}$ is obtained by setting $m_0 = 30 \sim 60$, together with a value of $N = 2^{10}$. For short monitoring frequency $\Delta_t < 1/200$, we recommend a value of $N = 2^{11}$ to obtain practical accuracy of several decimals, given

that the transition densities become more peaked with decreasing Δ_t . Finally, a large set of Bermudan reference prices are recorded in Table 18 for each model across a set of strikes. Note that time to maturity T and M varies for each set of Bermudan prices.

m_0	$M = 12$				$M = 52$				$M = 252$			
	10	20	40	60	10	20	40	60	10	20	40	60
9	0.11	0.30	0.85	1.53	0.16	0.54	1.68	3.35	0.44	1.61	5.96	12.56
$\log_2(N)$ 10	0.22	0.58	1.54	2.88	0.31	0.91	2.82	5.74	0.74	2.60	9.34	20.31
11	0.52	1.47	4.13	8.21	0.58	1.70	5.18	10.57	1.30	4.51	16.10	34.92

Table 17: CPU times in seconds for Bermudan option pricing in stochastic volatility models averaged over 20 trials each. Total log return space grid budget is N .

6 Conclusion

In this paper, we consider the problem of pricing American and barrier options under regime switching jump diffusion models and general stochastic volatility models with jumps. By working in the Fourier domain, we develop a novel, fast and accurate method for pricing generalized American and barrier options, including both discretely and continuously monitored contracts. Numerical experiments confirm the accuracy and efficiency of the proposed method. It would be interesting to extend the method to the case of American-style Asian options. Another future research direction is an extension to multi-factor stochastic volatility models (e.g. see Pun et al. (2015)) as well as the case of jumps in the variance process as in Salmi et al. (2014).

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References

- Ballestra, L. V. and G. Pacelli (2013). Pricing European and American options with two stochastic factors: A highly efficient radial basis function approach. *Journal of Economic Dynamics and Control* 37(6), 1142–1167.
- Barracough, K. and R. E. Whaley (2012). Early exercise of put options on stocks. *The Journal of Finance* 67(4), 1423–1456.
- Bates, D. S. (1996). Jumps and stochastic volatility: Exchange rate processes implicit in deutsche mark options. *Review of Financial Studies* 9(1), 69–107.
- Battauz, A., M. De Donno, and A. Sbuelz (2014). Real options and american derivatives: The double continuation region. *Management Science* 61(5), 1094–1107.

K	Test Set 1					Test Set 2				
	90	95	100	105	110	90	95	100	105	110
No Jumps: $T = 0.5, M = 50$										
SS	1.1225	2.1520	3.8382	6.4099	10.1181	1.8752	3.1442	5.0023	7.5775	10.9557
HW	1.3635	2.6666	4.6546	7.4028	10.9155	0.9106	2.0446	3.9392	6.7229	10.4221
3/2	0.8844	2.0153	3.9216	6.7248	10.4363	1.1072	2.3386	4.3043	7.0918	10.6981
4/2	1.3647	2.6951	4.7245	7.5124	11.0396	1.3346	2.6513	4.6671	7.4462	10.9756
Hes	1.3270	2.5582	4.4700	7.1732	10.7062	1.4404	2.6739	4.5584	7.2115	10.6960
Scott	0.5990	1.5434	3.3137	6.1317	10.0860	0.9989	2.1084	3.9496	6.6782	10.3618
α -H	1.3081	2.5165	4.4025	7.0880	10.6271	0.9140	1.9557	3.7322	6.4411	10.2023
Normal Jumps: $T = 0.5, M = 20$										
SS	4.2406	5.6759	7.5326	9.9051	12.8623	4.7924	6.3802	8.3711	10.8225	13.7713
HW	4.4259	6.0044	8.0462	10.5965	13.6585	4.1042	5.5674	7.5166	10.0239	13.1030
3/2	4.0942	5.5517	7.5010	10.0153	13.1047	4.2487	5.7723	7.7782	10.3218	13.4055
4/2	4.4374	6.0284	8.0927	10.6716	13.7601	4.4112	5.9933	8.0475	10.6165	13.6975
Hes	4.3932	5.9315	7.9210	10.4213	13.4515	4.4724	6.0213	8.0071	10.4881	13.4887
Scott	3.8976	5.2448	7.0820	9.5251	12.6114	4.1619	5.6194	7.5412	10.0079	13.0494
α -H	4.3831	5.9124	7.8903	10.3794	13.4018	4.1025	5.5251	7.4053	9.8372	12.8647
Double Exponential Jumps: $T = 0.5, M = 50$										
SS	3.0018	4.2673	6.0129	8.3615	11.4197	3.6068	5.0492	6.9472	9.3777	12.3975
HW	3.1938	4.6279	6.5867	9.1325	12.2811	2.8448	4.1468	6.0022	8.5090	11.7034
3/2	2.8300	4.1256	5.9825	8.4995	11.7075	2.9982	4.3698	6.2900	8.8343	12.0201
4/2	3.2013	4.6502	6.6350	9.2127	12.3881	3.1747	4.6135	6.5873	9.1552	12.3250
Hes	3.1639	4.5522	6.4534	8.9475	12.0714	3.2521	4.6507	6.5446	9.0126	12.0984
Scott	2.6177	3.7861	5.5173	7.9635	11.2008	2.9105	4.2057	6.0314	8.4952	11.6527
α -H	3.1512	4.5270	6.4117	8.8895	12.0029	2.8457	4.0989	5.8757	8.3008	11.4521
Mixture of Normal Jumps: $T = 0.25, M = 20$										
SS	1.7562	2.8493	4.7188	7.5933	11.3881	2.2495	3.5833	5.5984	8.4000	11.9422
HW	1.9435	3.2686	5.3670	8.2836	11.9000	1.6981	2.8876	4.9264	7.8909	11.6158
3/2	1.6946	2.8823	4.9267	7.8989	11.6267	1.8115	3.0762	5.1579	8.1063	11.7756
4/2	1.9414	3.2768	5.3950	8.3292	11.9505	1.9330	3.2638	5.3776	8.3092	11.9311
Hes	1.9526	3.2343	5.2759	8.1592	11.7836	1.9948	3.2846	5.3176	8.1805	11.7870
Scott	1.5452	2.5923	4.5478	7.5553	11.3950	1.7704	2.9617	4.9652	7.8870	11.5926
α -H	1.8982	3.1610	5.1998	8.0977	11.7429	1.7094	2.8539	4.8264	7.7581	11.5041

Table 18: Bermudan options with stochastic volatility parameters from Table 12 and jump distribution parameters from Table 13. Other parameters: $S_0 = 100$, $r = 0.05$, and $q = 0.0$. Prices found with $N = 2^{13}$, $m_0 = 120$, and $\hat{\alpha} = 8$ for jump models, $\hat{\alpha} = 6$ for no jumps.

Broadie, M. and J. Detemple (2004). Option pricing: Valuation models and applications. *Management Science* 50, 1145–1177.

Broadie, M., J. Detemple, E. Ghysels, and O. Tores. (2000). American options with stochastic dividends and volatility: A nonparametric investigation. *Journal of Econometrics* 94, 5392.

Broadie, M. and P. Glasserman (1997). Pricing American-style securities by simulation. *Journal of Economics Dynamics and Control* 21, 1323–1352.

Buffington, J. and R. Elliot (2002). American options with regime switching models. *International Journal of Theoretical and Applied Finance* 05, 1–26.

Carr, P. (1998). Randomization and the American Put. *Review of Financial Studies* 11, 597–626.

- Carr, P., R. Jarrow, and R. Myneni (1992). Alternative characterizations of American put options. *Mathematical Finance* 2, 87–106.
- Carr, P. and J. Sun (2007). A new approach for option pricing under stochastic volatility. *Review of Derivatives Research* 10(2), 87–150.
- Carr, P. and L. Wu (2003). What type of process underlies options? a simple robust test. *The Journal of Finance* 58(6), 2581–2610.
- Chockalingam, A. and K. Muthuraman (2011). American Options Under Stochastic Volatility. *Operations Research* 59, 793–809.
- Copeland, T. and P. Tufano (2004). A real-world way to manage real options. *Harvard Business Review* 82(3), 90–99.
- Cox, J. C., S. A. Ross, and M. Rubinstein (1979). Option pricing: A simplified approach. *Journal of Financial Economics* 7, 229–263.
- Da Fonseca, J. and C. Martini (2016). The α -Hypergeometric stochastic volatility model. *Stochastic Processes and their Applications* 126(5), 1472–1502.
- Detemple, J. (2005). *American-style derivatives: Valuation and computation*. CRC Press.
- Detemple, J. and T. Emmerling (2009). American chooser options. *Journal of Economic Dynamics and Control* 33(1), 128–153.
- Detemple, J. and W. Tian (2002). The valuation of American options for a class of diffusion processes. *Management Science* 48(7), 917–937.
- Duffie, D., J. Pan, and K. Singleton (2000). Transform analysis and asset pricing for affine jump-diffusions. *Econometrica* 68(6), 1343–1376.
- Fang, F. and C. Oosterlee (2008). A novel pricing method for European options based on Fourier cosine series expansions. *SIAM Journal on Scientific Computing* 31, 826–848.
- Fang, F. and C. Oosterlee (2009). Pricing early-exercise and discrete barrier options by Fourier-cosine series expansions. *Numerische Mathematik* 114, 27–62.
- Fang, F. and C. Oosterlee (2011). A Fourier-based valuation method for Bermudan and barrier options under Heston’s model. *SIAM J. Finan. Math.* 2, 439–463.
- Feng, L. and X. Lin (2013). Pricing Bermudan options in Levy process models. *SIAM J. Finan. Math.* 4(1), 474–493.
- Florescu, I., R. Liu, M. Mariani, and G. Sewell (2013). Numerical schemes for option pricing in regime-switching jump diffusion models. *International Journal of Theoretical and Applied Finance* 16, 1–25.
- Fujiwara, H. and M. Kijima (2007). Pricing of path-dependent american options by monte carlo simulation. *Journal of Economic Dynamics and Control* 31(11), 3478–3502.
- Gihman, I. I. and A. V. Skorohod (1979). Stochastic differential equations. In *The Theory of Stochastic Processes III*, pp. 113–219. Springer.
- Grasselli, M. (2016). The 4/2 Stochastic Volatility Model: a unified approach for the Heston and the 3/2 model. *Mathematical Finance*, forthcoming .
- Heston, S. (1993). A closed-form solution for option pricing with stochastic volatility with application to bond and currency options. *The Review of Financial Studies* 6, 327–343.

- Hull, J. and A. White (1990). Pricing interest-rate derivative securities. *The Review of Financial Studies* 3, 735–792.
- Ikonen, S. and J. Toivanen (2004). Operator splitting methods for American option pricing. *Applied Mathematics Letters* 17, 809–814.
- Ikonen, S. and J. Toivanen (2008). Efficient numerical methods for pricing American options under stochastic volatility. *Numerical Methods for Partial Differential Equations* 24(1), 104–126.
- Jacka, S. (1991). Optimal stopping and the American put. *Mathematical Finance* 1, 1–14.
- Jensen, M. V. and L. H. Pedersen (2016). Early option exercise: Never say never. *Journal of Financial Economics*, forthcoming .
- Jiang, J., R. Liu, and D. Nguyen (2016). A recombining tree method for option pricing with state-dependent switching rates. *International Journal of Theoretical and Applied Finance* 19, 1–26.
- Khaliq, A. Q. M. and R. Liu (2009). New numerical scheme for pricing American option under regime switching model. *International Journal of Theoretical and Applied Finance* 12(3), 319–340.
- Kirkby, J. (2015). Efficient option pricing by frame duality with the fast Fourier transform. *SIAM Journal on Financial Mathematics* 6(1), 713–747.
- Kirkby, J. (2016). An efficient transform method for Asian option pricing. *SIAM Journal on Financial Mathematics* 7(1), 845–892.
- Kirkby, J. (2017a). American and exotic option pricing with jump diffusions and other Lévy processes. Working Paper.
- Kirkby, J. (2017b). Robust barrier option pricing by frame projection under exponential Levy dynamics. *Applied Mathematical Finance* 24(4), 337–386.
- Kou, S. (2002). A jump-distribution model for option pricing. *Management Science* 48, 1086–1101.
- Kushner, H. (1990). Numerical methods for stochastic control problems in continuous time. *SIAM Journal of Control and Optimization* 28, 999–1048.
- Lee, S. S. and J. Hannig (2010). Detecting jumps from Lévy jump diffusion processes. *Journal of Financial Economics* 96(2), 271–290.
- Lindset, S. and A.-C. Lund (2007). A monte carlo approach for the american put under stochastic interest rates. *Journal of Economic Dynamics and Control* 31(4), 1081–1105.
- Liu, R. and D. Nguyen (2015). A tree approach to options pricing under regime-switching jump diffusion models. *International Journal of Computer Mathematics* 92(12), 2575–2595.
- Lo, C. and K. Skindilias (2014). An improved Markov chain approximation methodology: Derivatives pricing and model calibration. *International Journal of Theoretical and Applied Finance* 17, 407–446.
- Longstaff, F. and E. S. Schwartz (2001). Valuing American options by simulation: Simple least-squares approach. *Review of Financial Studies* 14, 113–147.
- Lord, R., F. Fang, F. Bervoets, and C. Oosterlee (2008). A fast and accurate FFT-based method for pricing early-exercise options under Levy processes. *SIAM J. Sci. Comput.* 10, 1678–1705.
- Medvedev, A. and O. Scaillet (2010). Pricing american options under stochastic volatility and stochastic interest rates. *Journal of Financial Economics* 98(1), 145–159.

- Merton, R. C. (1976). Option pricing when underlying stock returns are discontinuous. *Journal of financial economics* 3(1-2), 125–144.
- Muthuraman, K. (2008). A moving boundary approach to American option pricing. *Journal of Economics Dynamics and Control* 32, 3520–3537.
- Myneni, R. (1992). The pricing of the American option. *Annal of Applied Probability* 2, 1–23.
- Pun, C. S., S. F. Chung, and H. Y. Wong (2015). Variance swap with mean reversion, multifactor stochastic volatility and jumps. *European Journal of Operational Research* 245(2), 571–580.
- Rambeericha, N. and A. A. Pantelous (2016). A high order finite element scheme for pricing options under regime switching jump diffusion processes. *Journal of Computational and Applied Mathematics* 300, 83–96.
- Rambharat, B. R. and A. E. Brockwell (2010). Sequential Monte Carlo pricing of American-style options under stochastic volatility models. *The Annals of Applied Statistics* 4, 222–265.
- Rubinstein, M. (1994). Implied binomial trees. *The Journal of Finance* 49, 771–818.
- Salmi, S., J. Toivanen, and L. Sydow (2014). An IMEX-Scheme for Pricing Options under Stochastic Volatility Models with Jumps. *SIAM Journal of Scientific Computing* 36, 817–834.
- Scott, L. (1987). Option pricing when the variance changes randomly: Theory, estimation, and an application. *Journal of Financial and Quantitative Analysis* 22(4), 419–438.
- Stein, E. M. and J. Stein (1991). Stock price distributions with stochastic volatility: An analytic approach. *The Review of Financial Studies* 4, 272–752.
- Tavella, D. and C. Randall (2000). *Pricing Financial Instruments-The Finite Difference Method*. Wiley Series In Financial Engineering, John Wiley & Sons, New York.
- Zeng, P. and Y. Kwok (2014). Pricing barrier and Bermudan style options under time-changed Lévy processes: fast Hilbert transform approach. *SIAM Journal on Scientific Computing* 36(3), 450–485.
- Zhu, S.-P. (2006). An exact and explicit solution for the valuation of american put options. *Quantitative Finance* 6(3), 229–242.
- Zhu, S.-P. and W.-T. Chen (2013). An inverse finite element method for pricing american options. *Journal of Economic Dynamics and Control* 37(1), 231–250.
- Zhylyevskyy, O. (2010). A fast Fourier transform technique for pricing American options under stochastic volatility. *Review of Derivatives Research* 13, 1–24.
- Zvan, R., P. Forsyth, and K. Vetzalb (1998). Penalty methods for American options with stochastic volatility. *Journal of Computational and Applied Mathematics* 2, 199–218.

A Algorithms

This section provides algorithms for pricing Bermudan and barrier options under stochastic volatility (SV) and regime-switching (RS) jump diffusions. The algorithms are presented for the SV case, while the RS case follows by simply supplying the RS model’s generator Q and defining $\bar{\Lambda}_{j,k} \equiv 1$ in initialization Algorithm 1. The only other difference is that the final interpolation step in Algorithms 2 and 4 is not required for RS, since there is no volatility grid.

Algorithm 1 Stochastic Volatility Initialization

Fix Δ, N
 $a = 1/\Delta$; $\Delta_\xi = 2\pi a/N$; $\xi_n = (n-1)\Delta_\xi$, $n = 1, \dots, N$
Fix a variance grid $\{v_j\}_{j=1}^{m_0}$ and generator Q as in Section 2.3
 $\tilde{\mathcal{E}}(\xi_n) = \Delta_t (Q' + \text{diag}(\psi_1(\xi_n), \dots, \psi_{m_0}(\xi_n)))$, $n = 1, \dots, N$
 $\tilde{\mathcal{E}}(\xi_1) \leftarrow \exp(\tilde{\mathcal{E}}(\xi_1))$ (matrix exponential)
 $\bar{\Lambda}_{j,k} = \exp(i\Delta_\xi f(v_k, v_j))$, $j, k = 1 \dots, m_0$; $\Lambda \leftarrow \bar{\Lambda}$ (standard exponential)
for $n = 2, \dots, N$ **do**
 $\tilde{\mathcal{E}}(\xi_n) \leftarrow \exp(\tilde{\mathcal{E}}(\xi_n)) \circ \Lambda$ (matrix exponential)
 $\Lambda \leftarrow \Lambda \circ \bar{\Lambda}$
end for
 $\zeta_n = \exp(-i(1 - N/2)\Delta \cdot \xi_n) \cdot (\sin(\xi_n/(2a))/\xi_n)^2/(2 + \cos(\xi_n/a))$, $n = 2, \dots, N$
 $\Upsilon_{a,N}^{\Delta_t} = \exp(-r\Delta_t)32a^4/N$
for $j = 1, \dots, m_0$ **do**
 for $k = 1, \dots, m_0$ **do**
 $H_1^{j,k} \leftarrow \tilde{\mathcal{E}}_{k,j}(\xi_1)/(24a^2)$
 $H_n^{j,k} \leftarrow \tilde{\mathcal{E}}_{k,j}(\xi_n) \cdot \zeta_n$, $n = 2, \dots, N$
 $\{\bar{\beta}_n^{j,k}\}_{n=1}^N \leftarrow \Re\{\text{FFT}(\{H_n^{j,k}\}_{n=1}^N)\}$
 $\mu^{j,k} = \text{FFT}\left\{\Upsilon_{a,N}^{\Delta_t} \cdot (\bar{\beta}_{N/2}^{j,k}, \dots, \bar{\beta}_1^{j,k}, 0, \bar{\beta}_{N-1}^{j,k}, \dots, \bar{\beta}_{N/2+1}^{j,k})^\top\right\}$
 end for
end for

Algorithm 2 Down-and-out Barrier Option (put/call) - Stochastic Volatility

Fix $\hat{\alpha}, N$; determine x_1, Δ, n_0 as in Section 2.5.1
Call initialization Algorithm 1
Coefficient Matrix: $\theta_{n,j} \leftarrow 0$, $n = 1, \dots, N/2$, $j = 1, \dots, m_0$
Initialize first column $\theta_{n,1}$ with θ_n^M defined in equation (44) or (46) (put/call)
 $\mathcal{C}_{n,j} \leftarrow 0$, $n = 1, \dots, N/2$, $j = 1, \dots, m_0$
 $\tilde{\theta} \leftarrow \text{FFT}\{(\theta_{1,1}, \dots, \theta_{N/2,1}, 0, \dots, 0)\}$
for $j = 1, \dots, m_0$ **do**
 for $k = 1, \dots, m_0$ **do**
 $p \leftarrow \text{iFFT}\{\mu^{j,k} \circ \tilde{\theta}\}$; $\{\mathcal{C}_{n,j}\}_{n=1}^{N/2} \leftarrow \{\mathcal{C}_{n,j}\}_{n=1}^{N/2} + \{p\}_{n=1}^{N/2}$
 end for
end for
for $m = M-2, \dots, 0$ **do**
 for $k = 1, \dots, m_0$ **do**
 $\theta_{1,k} = (13\mathcal{C}_{1,k} + 15\mathcal{C}_{2,k} - 5\mathcal{C}_{3,k} + \mathcal{C}_{4,k})/48$
 $\theta_{n,k} = (\mathcal{C}_{n-1,k} + 10\mathcal{C}_{n,k} + \mathcal{C}_{n+1,k})/12$, $n = 2, \dots, N/2-1$
 $\theta_{N/2,k} = (13\mathcal{C}_{N/2,k} + 15\mathcal{C}_{N/2-1,k} - 5\mathcal{C}_{N/2-2,k} + \mathcal{C}_{N/2-3,k})/48$
 end for
 Call Algorithm 3 to update $\{\mathcal{C}_{n,j}\}$
end for
Find $k_0 \in \{1, \dots, m_0\}$ such that $v_{k_0} \leq v_0 < v_{k_0+1}$
Price(v_0) := $C_{n_0,k_0} + (C_{n_0,k_0+1} - C_{n_0,k_0}) \cdot (v_0 - v_{k_0})/(v_{k_0+1} - v_{k_0})$

Algorithm 3 Update Continuation Value

```
 $\mathcal{C}_{n,j} \leftarrow 0, \quad n = 1, \dots, N/2, \quad j = 1, \dots, m_0$ 
for  $k = 1, \dots, m_0$  do
   $\tilde{\theta} \leftarrow \text{FFT}\{(\theta_{1,k}, \dots, \theta_{N/2,k}, 0, \dots, 0)\}$ 
  for  $j = 1, \dots, m_0$  do
     $p \leftarrow \text{iFFT}\{\mu^{j,k} \circ \tilde{\theta}\}$ 
     $\{\mathcal{C}_{n,j}\}_{n=1}^{N/2} \leftarrow \{\mathcal{C}_{n,j}\}_{n=1}^{N/2} + \{p\}_{n=1}^{N/2}$ 
  end for
end for
```

Remark 10 Our grid choice for Bermudan options aligns both $\ln(K/S_0)$ and $\ln(S_0/S_0) = 0$ with grid points. With grid-width parameter $\hat{\alpha} > 0$ chosen, we initially define $\tilde{\Delta} = 2\hat{\alpha}/(N-1)$, and set $n_0 = N/4$. There are four cases to consider to determine Δ . First, if $S_0 = K$, then $\Delta := \tilde{\Delta}$. Second, if $0 < |\ln(K/S_0)| < \tilde{\Delta}$, then $\Delta := \tilde{\Delta}$ and cubic spline interpolation is used at the end of the algorithm to determine the price from values at grid points $\{n_0-2, n_0-1, n_0, n_0+1, n_0+2\}$. This is done for each of the two states which bracket the initial regime, from which the price for v_0 is found by a second interpolation. In the next two cases, we assume $|\ln(K/S_0)| > \tilde{\Delta}$, and define $\bar{n} = \lfloor \ln(K/S_0)/\tilde{\Delta} + N/4 \rfloor$. Third, if $K < S_0$, $\Delta := \ln(K/S_0)/(1 + \bar{n} - N/4)$, and redefine $\bar{n} \leftarrow \bar{n} + 1$. Fourth, if $K > S_0$, $\Delta := \ln(K/S_0)/(\bar{n} - N/4)$. After Δ is found, we set $a := 1/\Delta$, $x_1 := (1 - N/4)\Delta$, and determine the frequency grid as in equation (31).

Algorithm 4 Bermudan Put - Stochastic Volatility

```
Fix  $\hat{\alpha}, N$ ; determine  $x_1, \Delta$  as in Remark 10;  $n_0 = N/4$ ;  $\bar{n} := \lfloor (a \cdot (\log(K/S_0) - x_1) + 1) \rfloor$ 
Call initialization Algorithm 1
Coefficient Matrix:  $\theta_{n,j} \leftarrow 0, \quad n = 1, \dots, N/2, \quad j = 1, \dots, m_0$ 
Initialize  $\theta_n^M$  defined in equation (56), and  $G_n = G_{S_0}(x_n), \quad n = 1, \dots, \bar{n}$ 
 $\mathcal{C}_{n,j} \leftarrow 0, \quad n = 1, \dots, N/2, \quad j = 1, \dots, m_0$ 
 $\tilde{\theta} \leftarrow \text{FFT}\{(\theta_1^M, \dots, \theta_{N/2}^M, 0, \dots, 0)\}$ 
for  $j = 1, \dots, m_0$  do
  for  $k = 1, \dots, m_0$  do
     $p \leftarrow \text{iFFT}\{\mu^{j,k} \circ \tilde{\theta}\}; \quad \{\mathcal{C}_{n,j}\}_{n=1}^{N/2} \leftarrow \{\mathcal{C}_{n,j}\}_{n=1}^{N/2} + \{p\}_{n=1}^{N/2}$ 
  end for
end for
for  $m = M-2, \dots, 0$  do
  for  $j = 1, \dots, m_0$  do
    Find  $\hat{n}(j), x_{\hat{n}(j)}$  in (49), and  $\hat{x}(j)$  in (50) using  $\{\mathcal{C}_{n,j}\}_{n=1}^{N/2}$  and  $\{G_n\}_{n=1}^{\bar{n}}$ 
     $\bar{\rho} \leftarrow \hat{x}(j) - x_{\hat{n}(j)}, \quad \zeta \leftarrow a\bar{\rho}$ 
    Find  $\delta_0^{put}, \bar{\delta}_0^{put}, \delta_1^{put}, \bar{\delta}_1^{put}$  by Table 2, and  $\gamma_1, \dots, \gamma_6$  by equation (52)
    Update  $\{\theta_{n,j}\}_{n=1}^{N/2}$  with definition of  $\theta_{n,j}^m$  in equation (51)
  end for
  Call Algorithm 3 to update  $\{\mathcal{C}_{n,j}\}_{n=1}^{N/2}$ 
end for
Find  $k_0 \in \{1, \dots, m_0\}$  such that  $v_{k_0} \leq v_0 < v_{k_0+1}$ ;  $\tau := (v_0 - v_{k_0})/(v_{k_0+1} - v_{k_0})$ 
Price( $v_0$ ) :=  $\max\{\mathcal{C}_{n_0,k_0}, G_{n_0}\} + (\max\{\mathcal{C}_{n_0,k_0+1}, G_{n_0}\} - \max\{\mathcal{C}_{n_0,k_0}, G_{n_0}\}) \cdot \tau$ 
```
